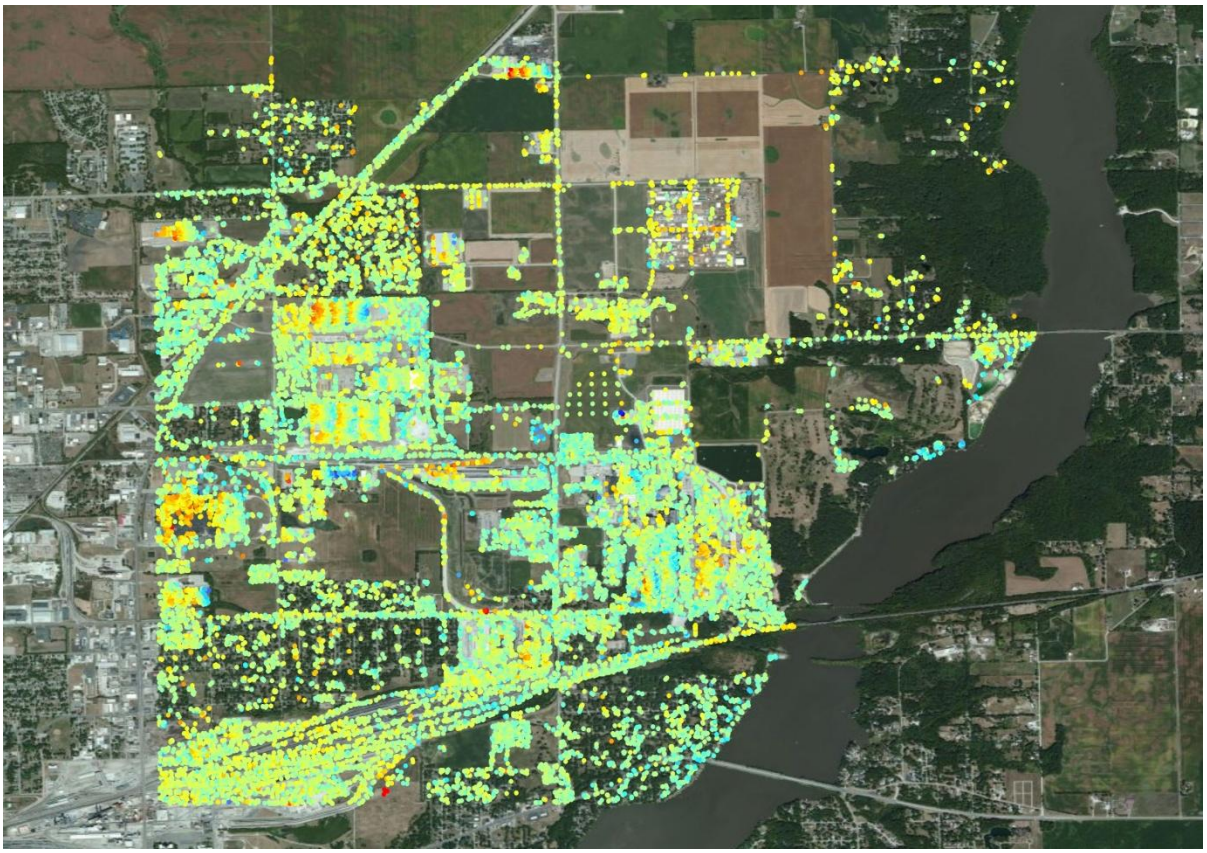
TRE®
Sensing the Planet

First Monitoring Report

InSAR Analysis of Ground Movement over the Decatur CO_2 Injection Site



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Executive Summary

This report describes the approach and results of the InSAR analysis being carried out to measure surface deformation at the Decatur injection site in response to CO_2 injection.

Imagery from the Cosmo-SkyMed satellite was acquired over the area of interest on a repeat acquisition frequency of approximately 8 days between 08 July 2011 and 31 December 2011. A total of 23 images were analyzed with TRE's proprietary SqueeSAR™ algorithm to identify measurement points from features within the area of interest, including a series of 21 artificial reflectors installed near the injection site. This report represents a baseline study over the Decatur site, as the majority of the images analyzed were acquired prior to the start of injection.

The deliverables for this stage of the project include this report, which describes the results obtained from the data processing, and a CD containing the displacement data.

The following points summarize the key findings of this analysis and the features of this deliverable:

- Surface displacement was observed in the area of the injection pad but the correlation with injection activities is not clear.
- Many measurement points appear to display seasonal movement trends. These effects will become better characterized once a full year of imagery is acquired.
- The standard deviation rates associated with estimated displacement rates was low.
- All artificial reflectors are clearly identifiable and appear to display deformation linked to seasonal trends. Any movement observed was small, and likely due to seasonal influences such as variations in soil moisture, temperature fluctuations or changes in ground water levels.
- From an InSAR perspective a high number of targets was identified (2,738 points per square mile - 1,057 points per square kilometer) which allows a comprehensive characterization of ground movement over the site.

Image acquisitions will continue over the Decatur site on an 8-day repeat schedule. The next data processing is scheduled for the beginning of April.

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1 Background

The CO_2 Capture Project (CCP) is a partnership of several of the world's leading energy companies with a mandate to research and develop technologies for the capture and geological storage of CO_2.

The Illinois Basin - Decatur Project is overseen by the Midwest Geological Sequestration Consortium (MGSC), and the site is being monitored with InSAR as an initial study on the use of InSAR in a North American environment. Supported by the CCP consortium, the objective is to demonstrate the effectiveness of advanced InSAR techniques, such as SqueeSAR™, in areas where agriculture, snowfall and the presence of natural vegetation will lower the density of radar targets.

With this objective TRE Canada Inc (TRE) performed an initial Feasibility Study in 2010 to identify the density and distribution of natural measurement points at the Decatur Test site. The outcome indicated a density of nearly 1,900 points per square mile; however, the distribution of point targets was non-uniform and low in non-urban areas. As a result, a series of artificial reflectors were installed in a vegetated area between the injection well and a nearby observation well to complement the natural reflectors.

This report describes the results obtained from the first stage of data processing over the Decatur injection site. The analysis was carried out using the SqueeSAR™ algorithm coupled with 6 months of COSMO-SkyMed X-band imagery.

2 Overview of the Area of Interest

The area of interest (AOI) is located northeast of the city of Decatur, Illinois and covers an extent approximately 9.75 square miles (25.25 km²) in size (Figure 1). Land cover within the AOI is varied and primarily consists of industrial sites, farmland, forest, and residential areas. The topography is mainly flat or gently sloping toward the lake.

This type of mixed environment resulted in a variable spatial coverage of measurement points being identified throughout the AOI. Many targets were identified from man-made structures and areas of exposed ground or fallow fields. In contrast, forested areas or areas of active agriculture produced a much lower density of radar targets due to the decorrelation of the radar signal over time.

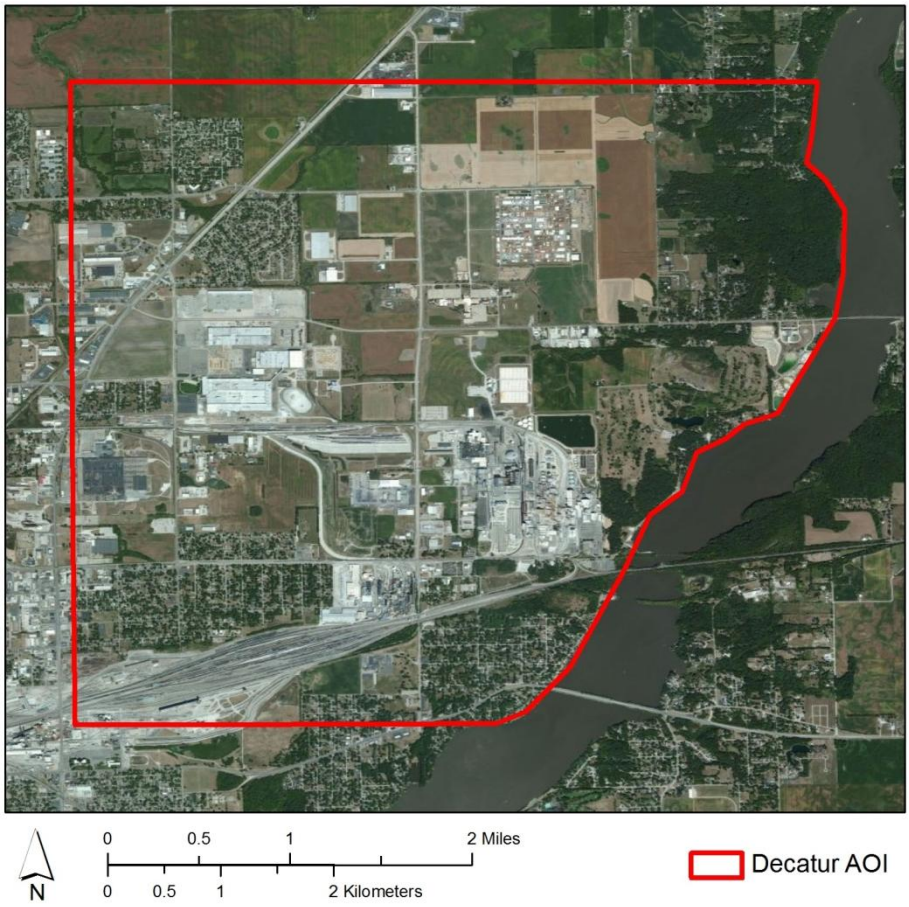


Figure 1. The area of interest (AOI) over the Decatur field.

Twenty-one artificial reflectors (AR) were installed at the Decatur site by the Illinois Geological Survey in order to provide guaranteed reflectors for the radar signal (Figure 2). The ARs are double geometry structures, meaning they are visible from both ascending and descending satellite orbits. The reflectors are based on a design provided by TRE, and can be viewed by all current satellites.

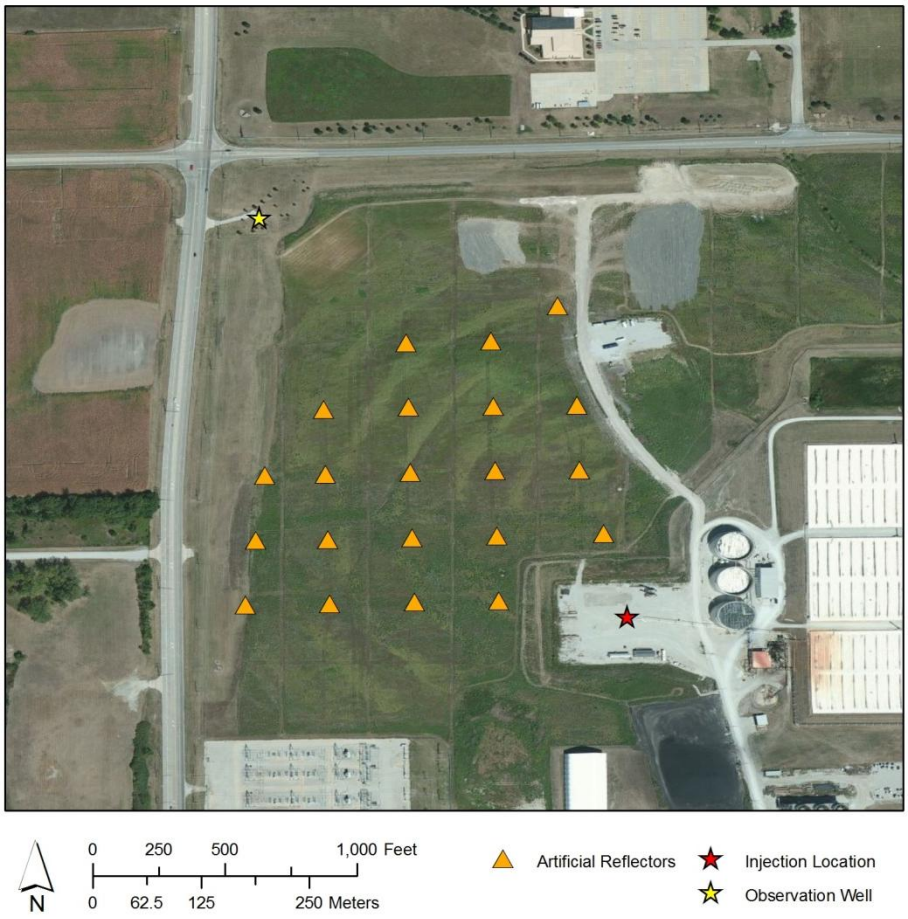


Figure 2. A close up of the Artificial Reflectors at the Decatur injection site.

3 Radar data

Consisting of four satellites, COSMO-SkyMed (CSK) radar imagery possesses a fine spatial resolution (3 x 3 m) and short repeat-pass interval (8 days). The use of a constellation of synchronized satellites allows for data redundancy, meaning missed acquisitions can be made up for by activating another satellite within the CSK constellation.

A total of 23 images were processed for the present report (Table 1). All images were acquired using beam mode H4-05 along an ascending orbit (meaning the satellite is travelling from south to north and imaging to the east). The data processed for the present report does not include the 4 images acquired for the visibility check (15 Mar 2010, and images acquired between 23 April 2011 to 13 May 2011). These images were not processed in order to avoid gaps in the dataset.

Additional information on satellite acquisition parameters can be found in Appendix 1, and further details on the InSAR processing are included in Appendix 3.

CSK Images	
1	7/8/2011
2	7/12/2011
3	7/16/2011
4	7/24/2011
5	8/1/2011
6	8/9/2011
7	8/17/2011
8	8/25/2011
9	9/2/2011
10	9/10/2011
11	9/26/2011
12	10/4/2011
13	10/12/2011
14	10/20/2011
15	10/28/2011
16	11/1/2011
17	11/13/2011
18	11/17/2011
19	11/21/2011
20	11/29/2011
21	12/15/2011
22	12/19/2011
23	12/31/2011

Table 1. Dates of the acquired CSK radar images.

4 Results of the InSAR analysis

4.1 Reference point

All measurements of displacement are made relative to a reference point that is assumed to be stable (Figure 3). The reference point is selected using an optimization procedure within the SqueeSAR™ processing chain. This approach involves an intensive statistical analysis of all of the identified targets in order to select a PS or DS with optimal radar parameters as the reference point (including high coherence, low standard deviation values and low variability over time). A neighborhood comparison of surrounding points is also performed to ensure the final selection and placement of the reference point is within a stable area. The use of an optimal reference point ensures the highest quality results are achieved, and allows for the best representation of surface deformation trends.

The coordinates of the reference point are listed in Appendix 1.

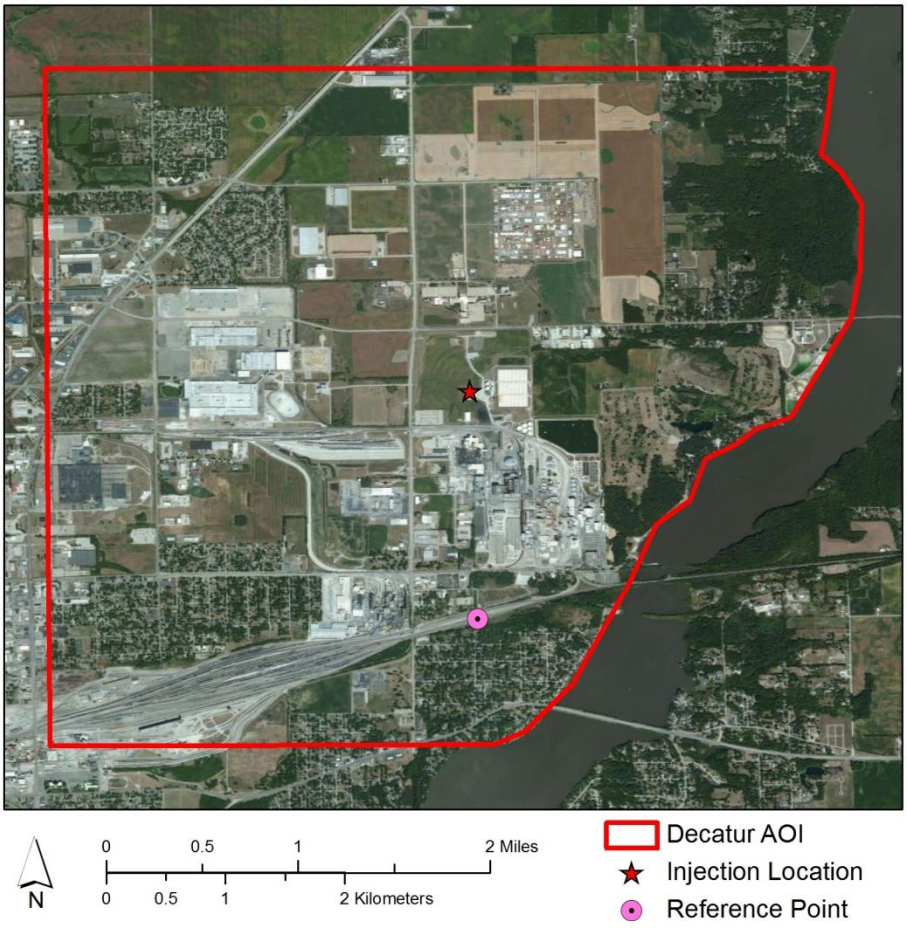


Figure 3. The location of the reference point.

4.2 Displacement rate

The line-of-sight (LOS) displacement rates, expressed in mm/year, as detected from the processing of the dataset is shown below (Figure 4). Each point corresponds to a Permanent Scatterer (PS) or a Distributed Scatterer (DS), and is color-coded according to its annual rate of movement. Average displacement values are calculated from a linear regression of the ground movement measured over the entire period covered by the satellite images. Detailed information on ground motion is also provided by means of displacement time series, which are provided for each PS and DS.

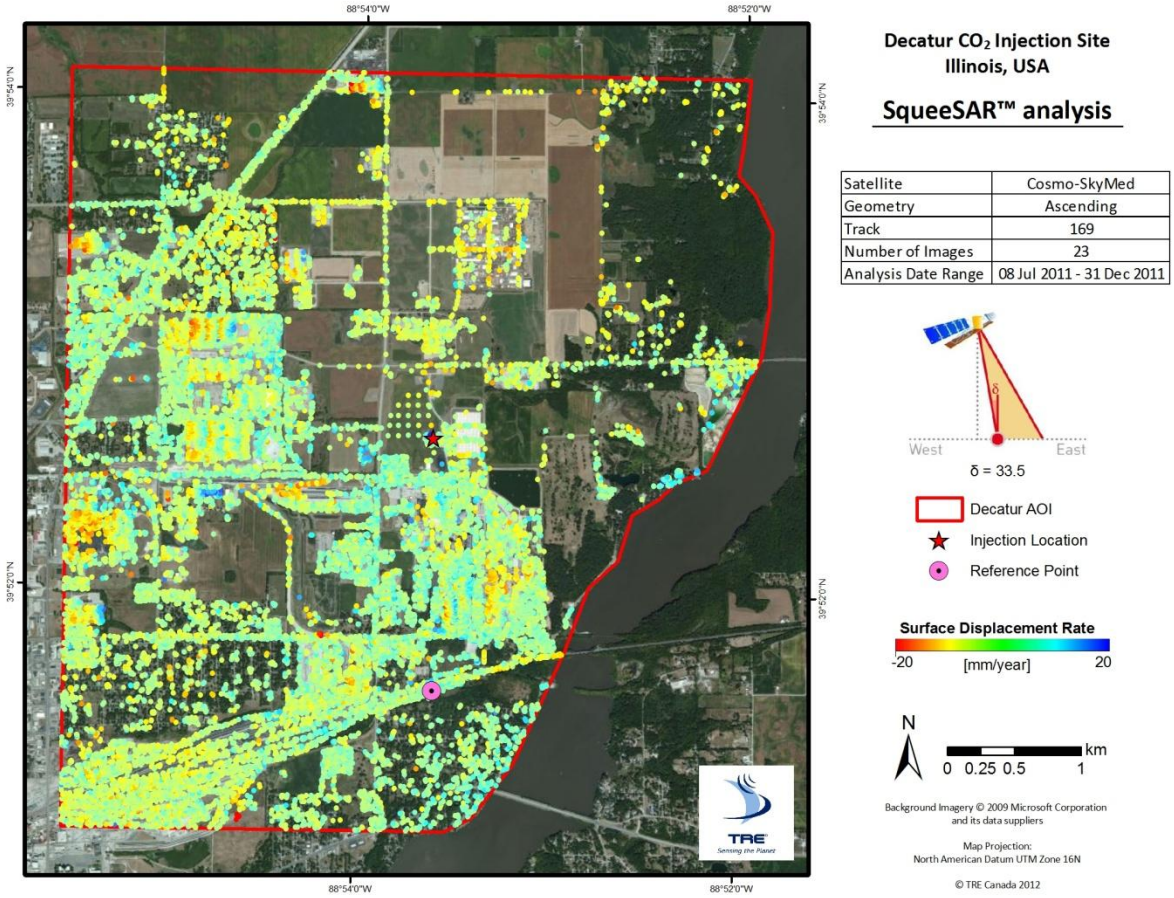


Figure 4. PS and DS deformation rates in mm/year.

4.3 Displacement Rate Standard Deviation

The standard deviation of the surface displacement data characterizes the error of the measurements (Figure 5). The displacement rate for a given point should be read in the form of *Displacement Rate* ± *Standard Deviation*. The precision of the displacement measurements are a function of the factors listed below and of local ground movement dynamics.

- Spatial density of the PS and DS (higher densities produce higher precisions)
- Quality of the radar targets (signal-to-noise ratio levels)
- Distance from the reference point
- Number of images processed
- Period of time covered by the imagery
- Climatic conditions at the time of the acquisitions
- Distance between the measurement point and the reference

The precision of the data (one standard deviation) is shown below (Figure 5). Additional information on data precision can be found in Appendix 1.

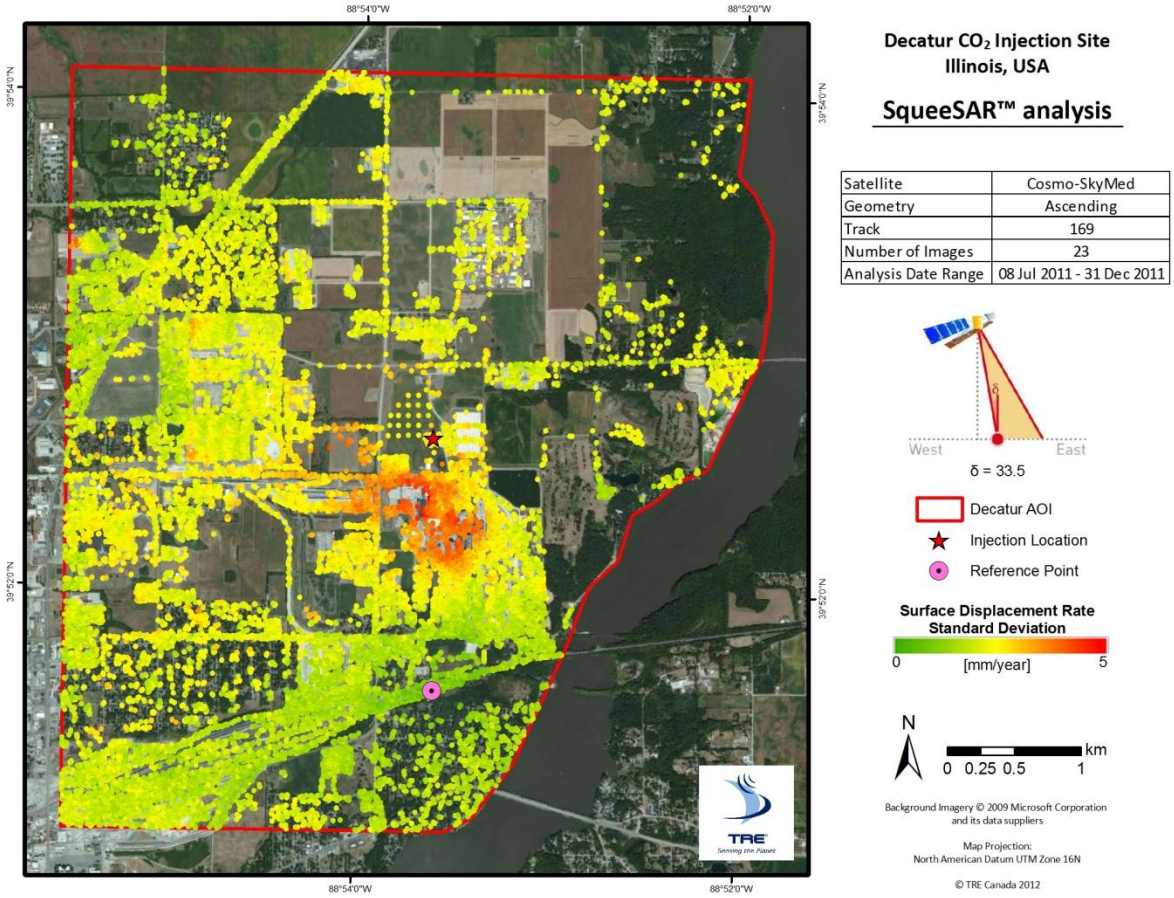


Figure 5. Standard deviation of the deformation rates.

4.4 Acceleration

PS and DS acceleration values (Figure 6) can be used to identify non-linear trends in the deformation time series. Acceleration is used to identify areas where the deformation rate is increasing or decreasing over time.

Negative accelerations are marked in red and indicate either an increase in downward movement rates, or a decrease in uplift. Positive accelerations are blue and indicate either an increase in the rate of uplift or a decrease in the rate of subsidence.

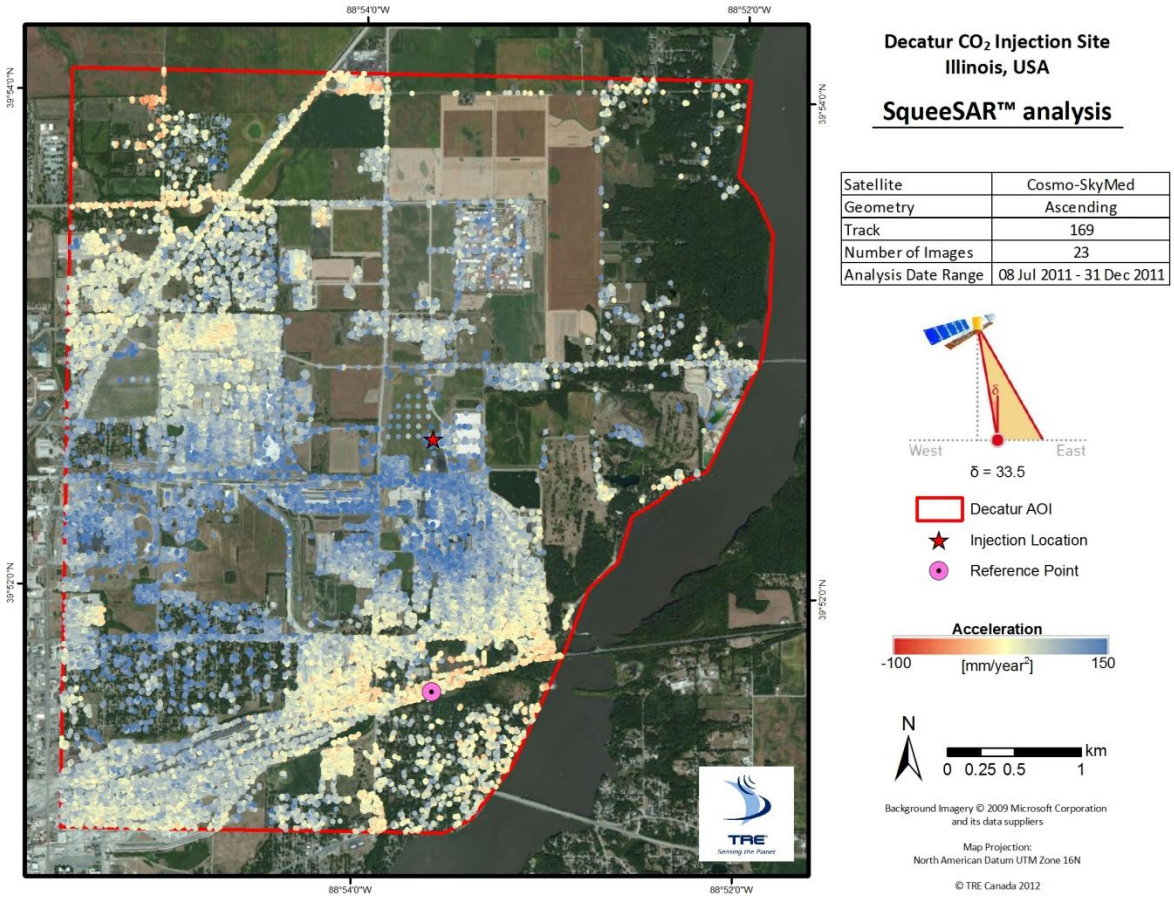


Figure 6. PS and DS acceleration values, expressed in mm/yr^2.

4.5 Acceleration Standard Deviation

The precision of the acceleration data (one standard deviation) is shown below (Figure 7). Standard deviation values of the estimated acceleration values are a function of the same factors impacting the precision of the displacement rates (see Section 4.3).

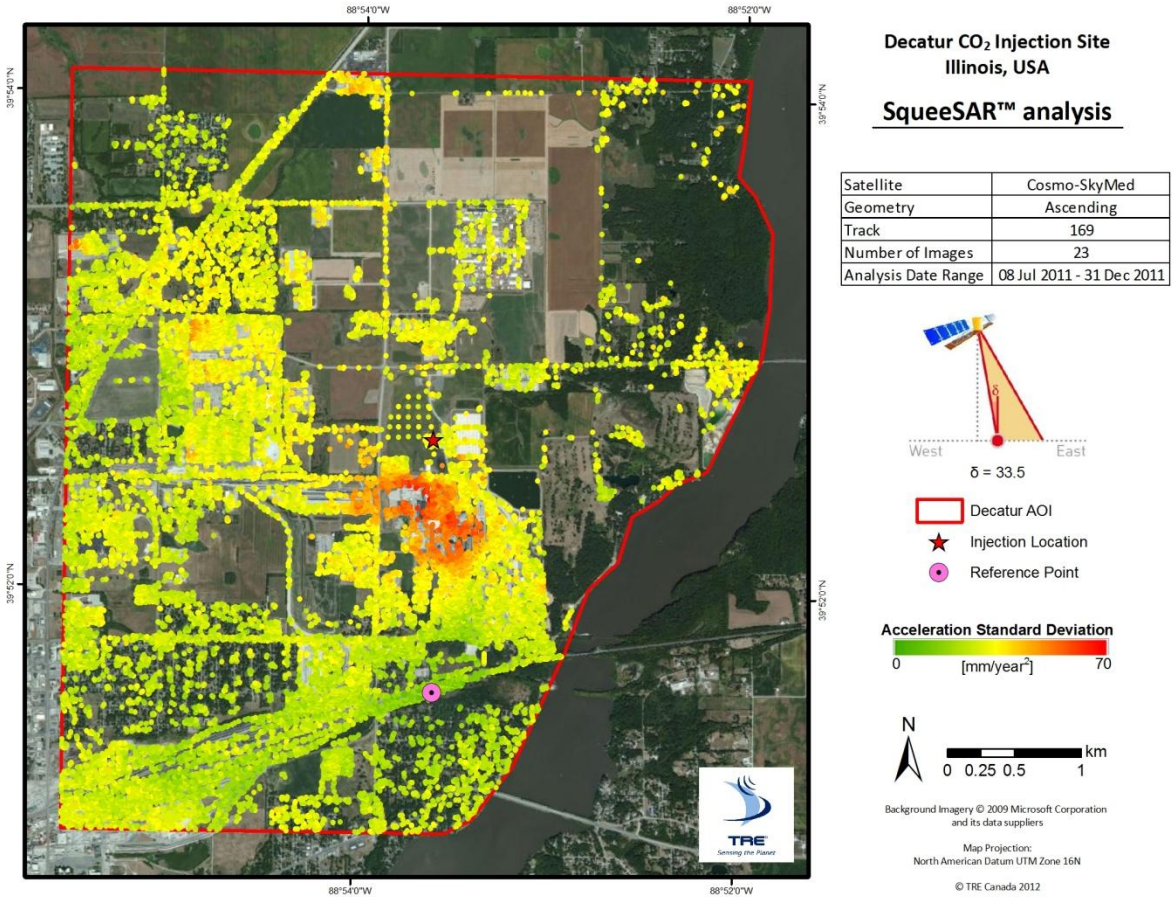


Figure 7. Standard deviation of the acceleration values, in mm/yr^2.

5 Observations

5.1 Overview of ground deformation

As many of the images used in this analysis were acquired prior to the start of injection, this report represents a baseline study of the ground deformation in the area, allowing for any pre-existing movement trends to be identified. While slight amounts of displacement were detected throughout the area, most of the observed movement appears to be related to seasonal trends. Ideally, one full year of data should be acquired before the start of injection in order to properly characterize seasonal movement in a baseline study; however, useful information can still be obtained from shorter data stacks. Seasonal movement will continue to be characterized over time with continued image acquisitions, and any movement related to injection activities will more clearly emerge.

5.2 Ground displacement at injection site

Injection of CO_2 began in November 2011. Several measurement points located on or close to the injection well pad show uplift (Figure 8); however, the start of deformation appears to pre-date the start of injection. The cause of this distinctive uplift pattern is uncertain and may be related to other site activities that TRE is not aware of. Time series for several of these points, as indicated below (Figure 8), are shown in the following section (Figure 10 to Figure 12, inclusive).

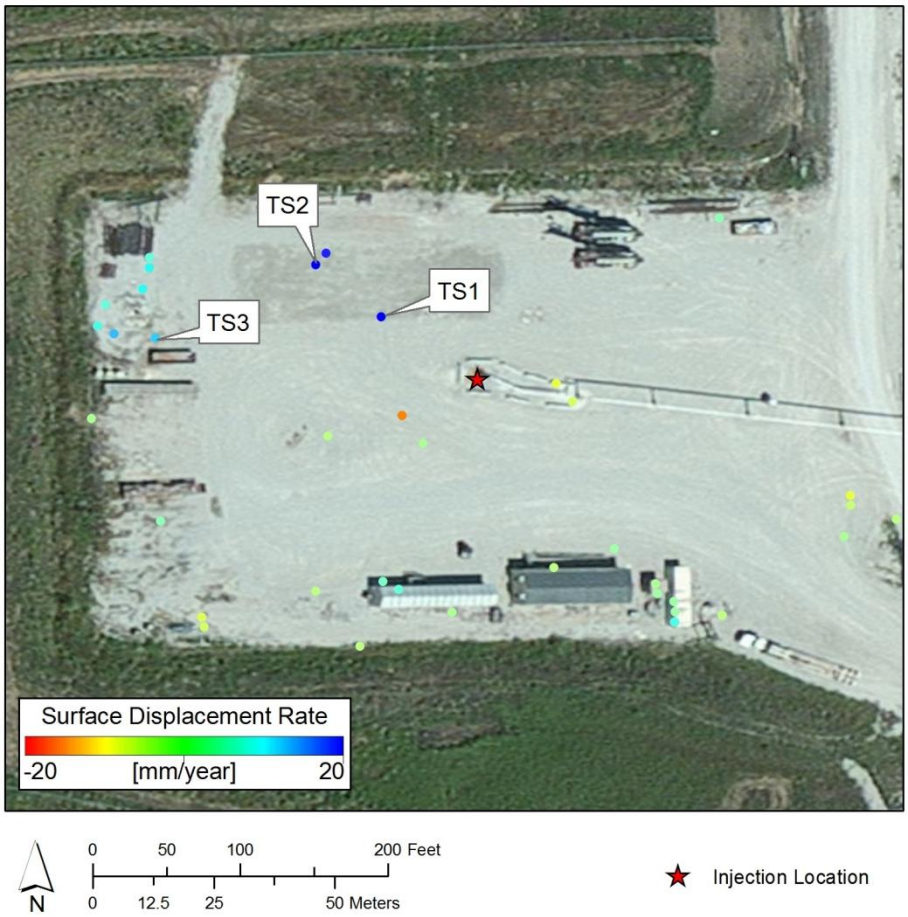


Figure 8. Measurement points displaying uplift in close proximity to the site of CO_2 injection at the Decatur field.

All of the measurement points with time series shown below (Figure 10 to Figure 12), are of the distributed scatterer type, meaning they do not represent displacement measured from a point-source. Distributed Scatterers represent movement over large homogeneous and contiguous areas of reflectivity (such as bare fields and exposed areas including pavement, building roofs, etc.). While DS are represented as individual points for clarity of presentation and ease of interpretation (e.g. Figure 8), these measurements actually correspond to non-point features that are multiple pixels in size.

Distributed scatterers that were identified over the injection site are shown below (Figure 9), symbolized by a square corresponding to the area of each DS. The square symbols represent DS size only, and do not correspond to the true shape/extent of the DS.

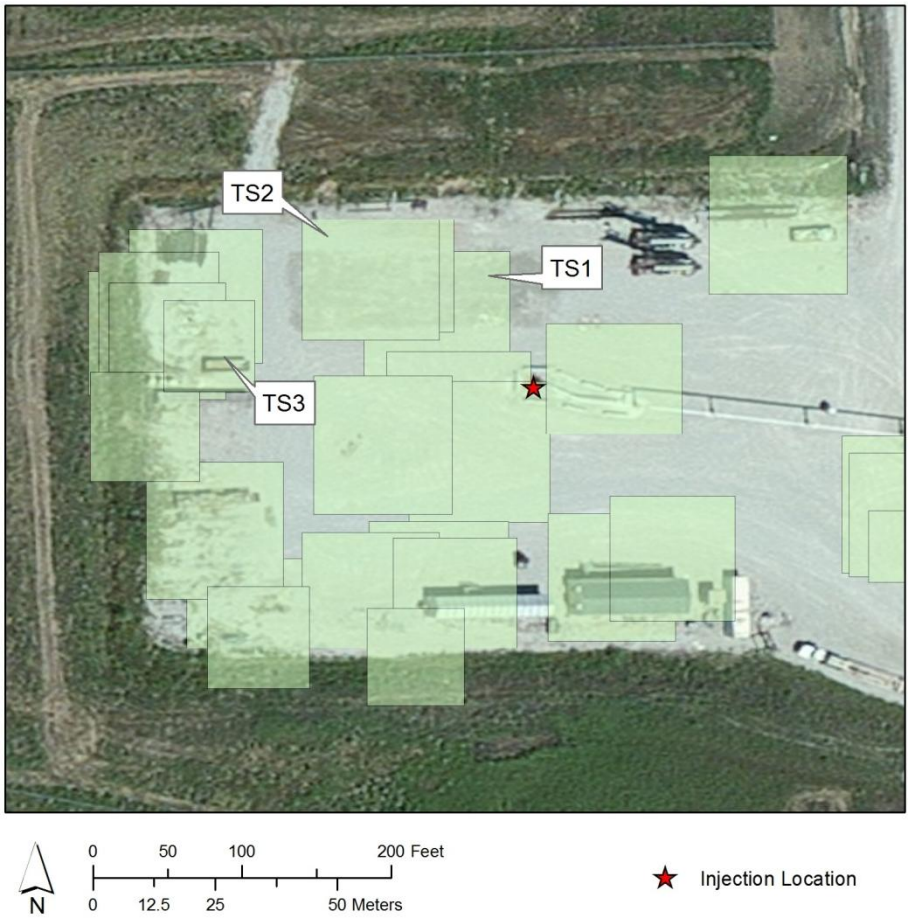


Figure 9. DS measurement points represented proportionally by the area they cover. (Note that the square does not represent the actual shape of the DS).

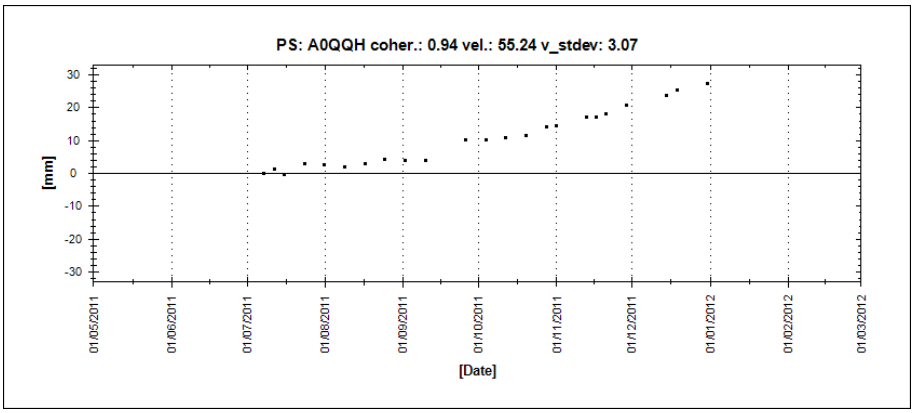


Figure 10. Time series for the point labeled TS1 in Figure 8.

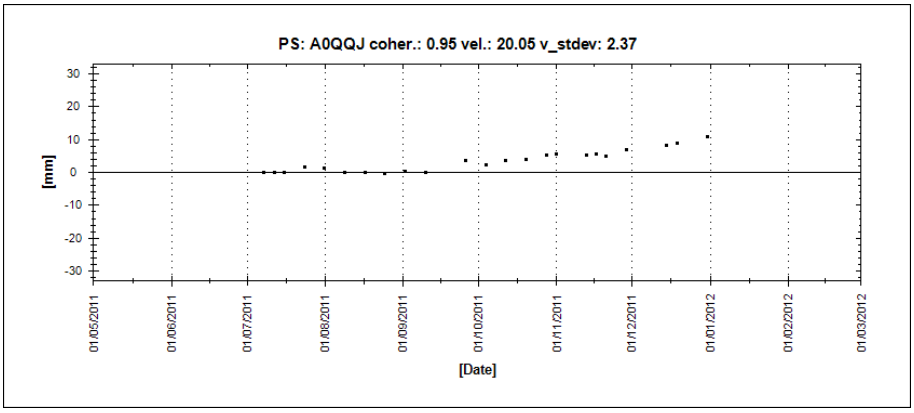


Figure 11. Time series for the point labeled TS2 in Figure 8.

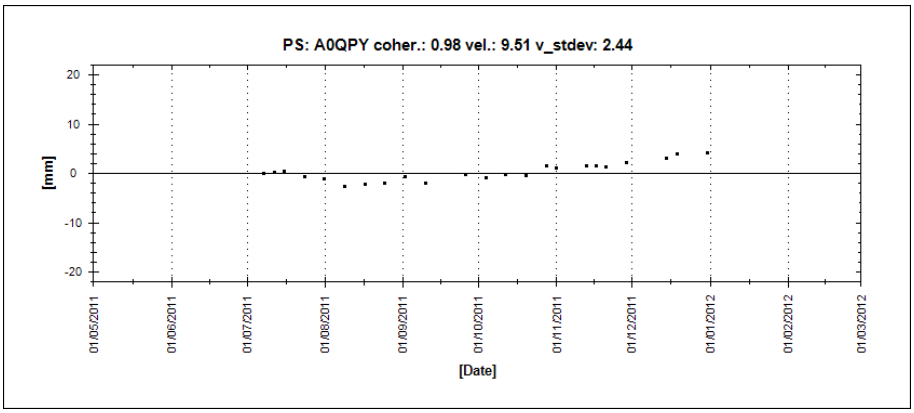


Figure 12. Time series for the point labeled TS3 in Figure 8.

5.3 Features of Interest

Two additional areas within the area analyzed demonstrated displacement trends that may be of interest (labelled as Feature 1 and 2 below), while a third (Feature 3) is highlighted due to higher standard deviation of the displacement measurements in this area (Figure 13). Each of these features are discussed in the following paragraphs.

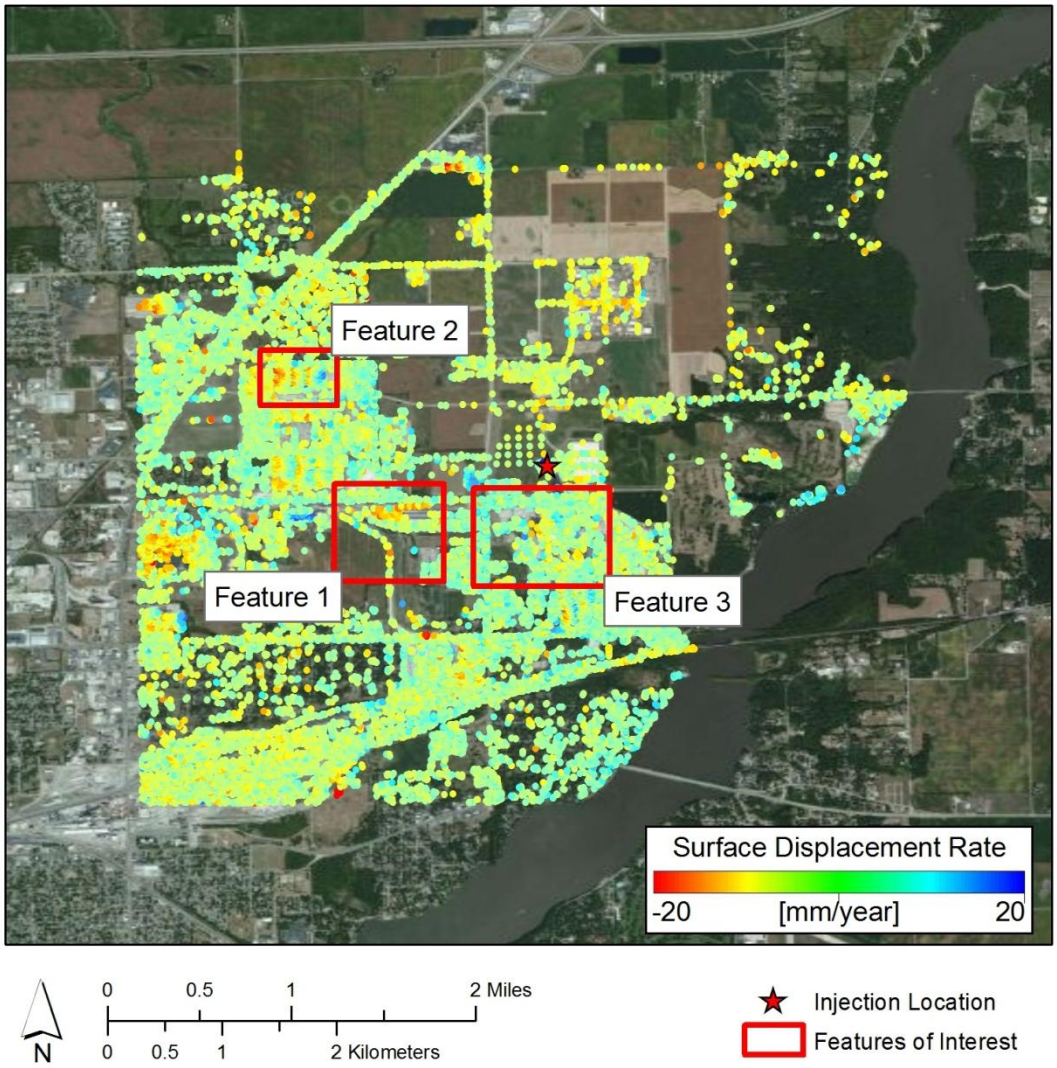


Figure 13. PS and DS displacement data over the Decatur area with the locations of three features of interest highlighted.

5.3.1 Feature 1

Subsidence was detected along a portion of the rail tracks located immediately southwest of the AOI (Figure 14). Time series for two of the points identified in this area are also shown (Figure 15 and Figure 16).

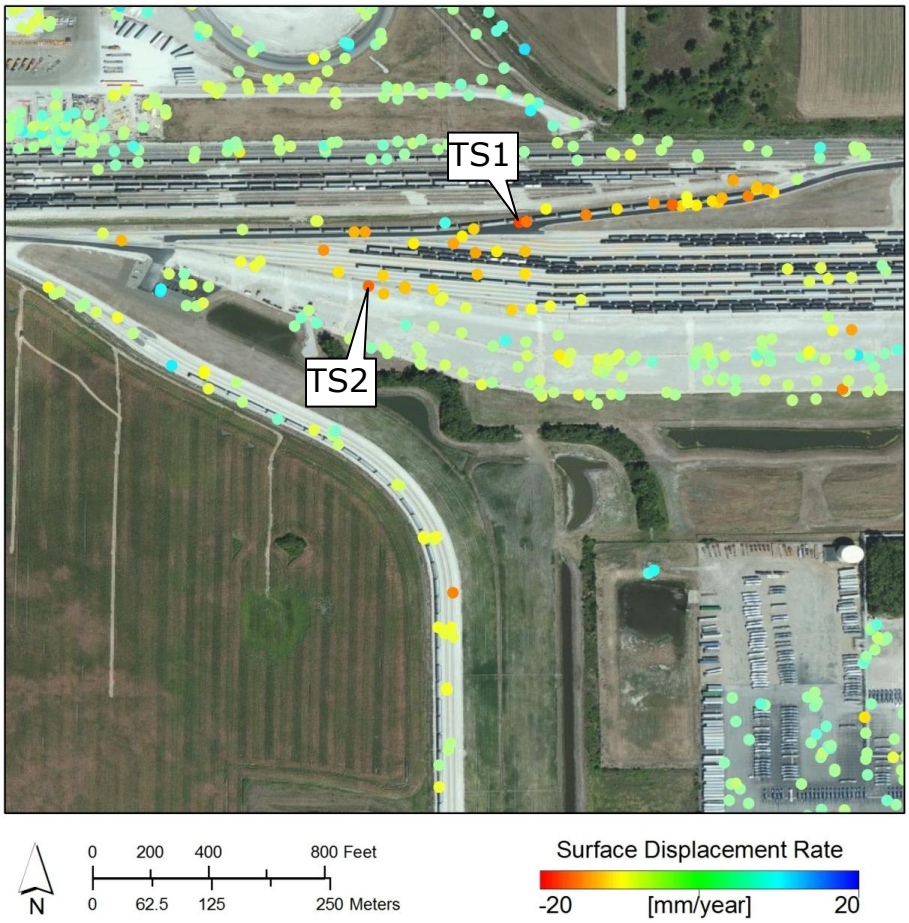


Figure 14. Surface displacement results for the area labelled Feature 1 in Figure 13.

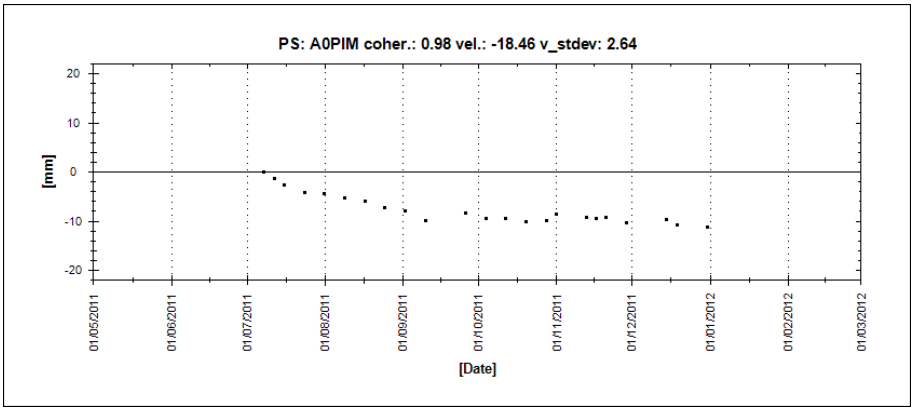


Figure 15. Time series for the point labelled TS1 in Figure 14.

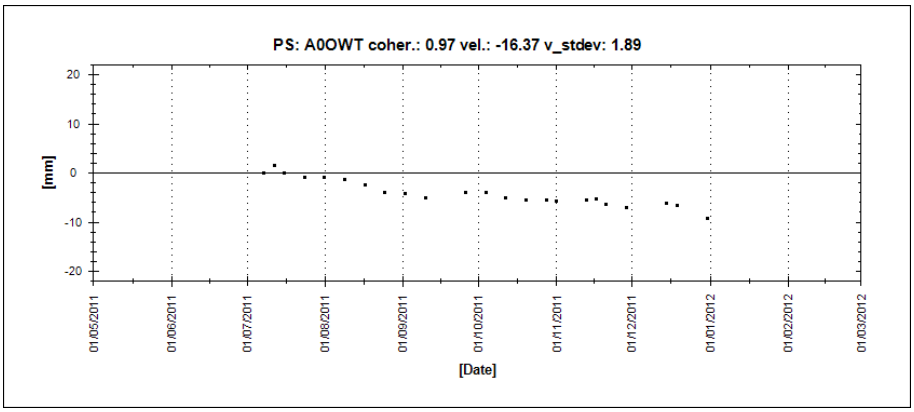


Figure 16. Time series for the point labelled TS2 in Figure 14.

5.3.2 Feature 2

Several buildings throughout the AOI appear to have a wide range of displacement rates across the length of the structure, resulting in the rainbow-like colouration when visualized according to their annual displacement rates. One example of this pattern is shown below (Figure 17). This displacement trend is likely caused by variations in the seasonal response of features across the roof of such buildings (e.g. differences in thermal contraction and expansion). This type of movement can be seen in the time series of the highlighted measurement points (Figure 18 to Figure 21).

As the annual displacement rate associated from each point is extrapolated from a dataset that covers 6 months, cyclical variations such as seasonal temperature fluctuations, can affect these shorter term rate estimations. Furthermore, as rates of movement are calculated as a linear regression of the measured data, these values are particularly sensitive to uplift or subsidence occurring near the end or beginning of a time series. It is expected that with the acquisition of one full year of data, seasonal variations will be better characterized and annual displacement rates will stabilize.

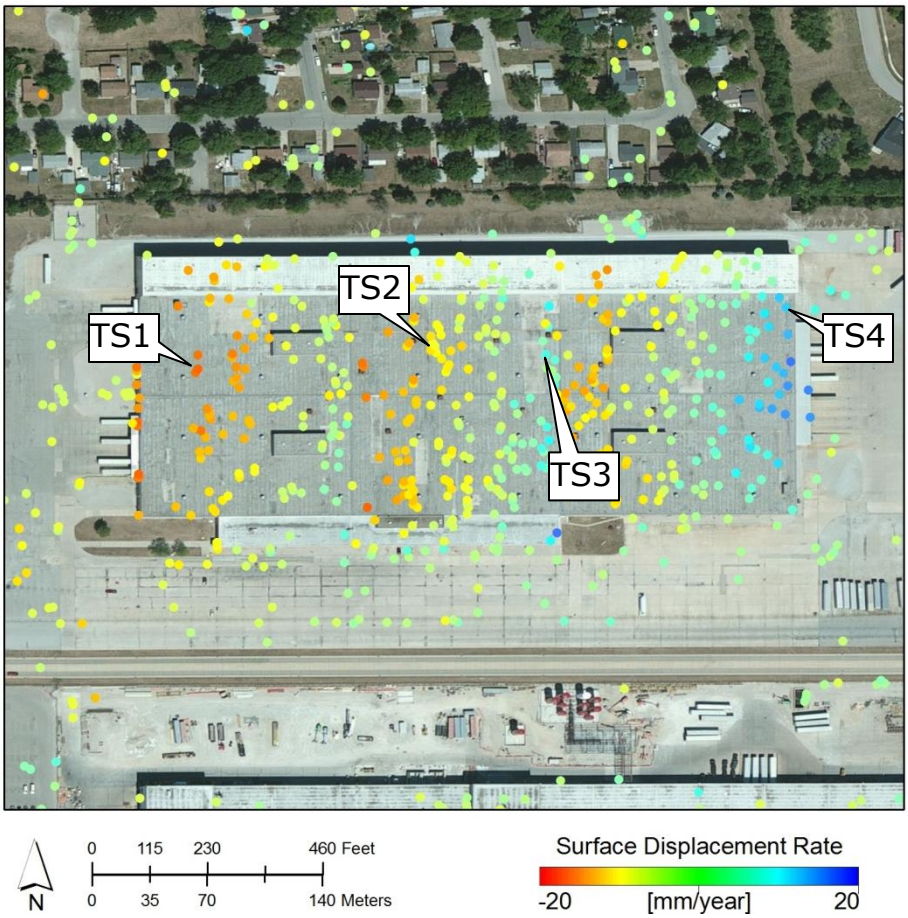


Figure 17. Surface displacement for the area labelled Feature 2 in Figure 13.

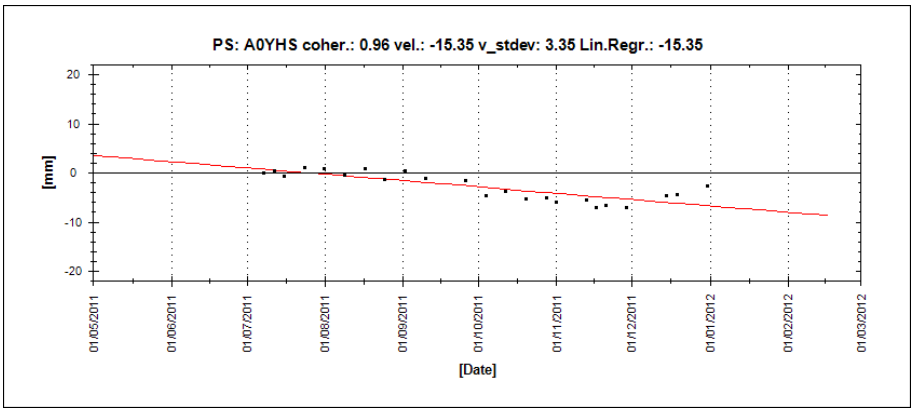


Figure 18. Time series for the point labelled TS1 in Figure 17. The red line represents the linear regression used to estimate the annual displacement rate.

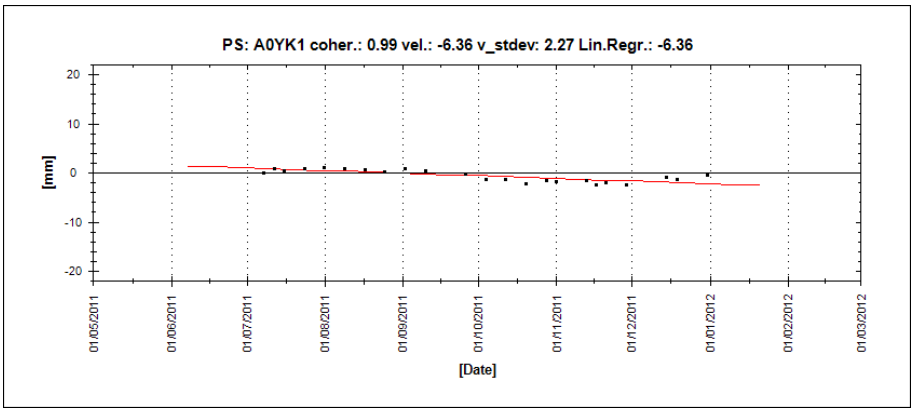


Figure 19. Time series for the point labelled TS2 in Figure 17. The red line represents the linear regression used to estimate the annual displacement rate.

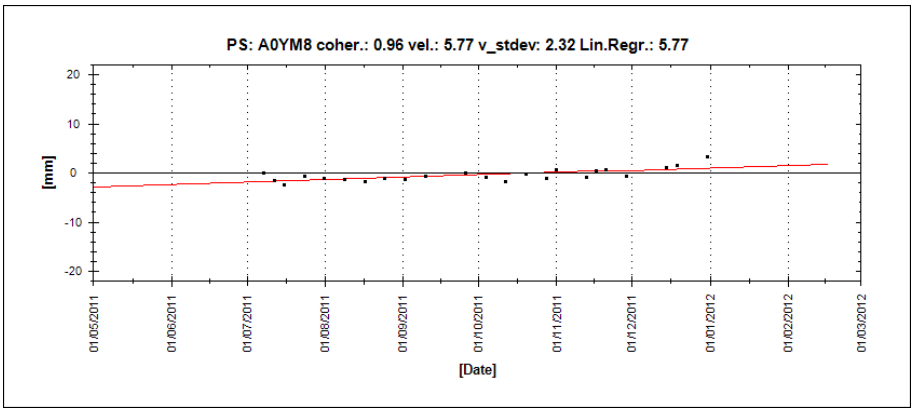


Figure 20. Time series for the point labelled TS3 in Figure 17. The red line represents the linear regression used to estimate the annual displacement rate.

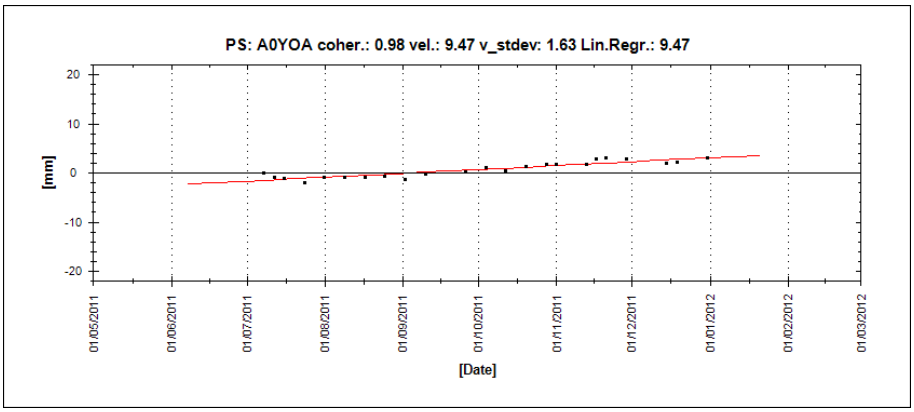


Figure 21. Time series for the point labelled TS4 in Figure 17. The red line represents the linear regression used to estimate the annual displacement rate.

5.3.3 Feature 3

The feature shown below is highlighted due to the comparatively high standard deviation rates associated with the measurement points in this area (Figure 22 and Figure 23). Time series for the two points highlighted below include the associated displacement and acceleration standard deviation values (Figure 24 and Figure 25). As more images are acquired it is expected that the standard deviation values in these areas will decrease.

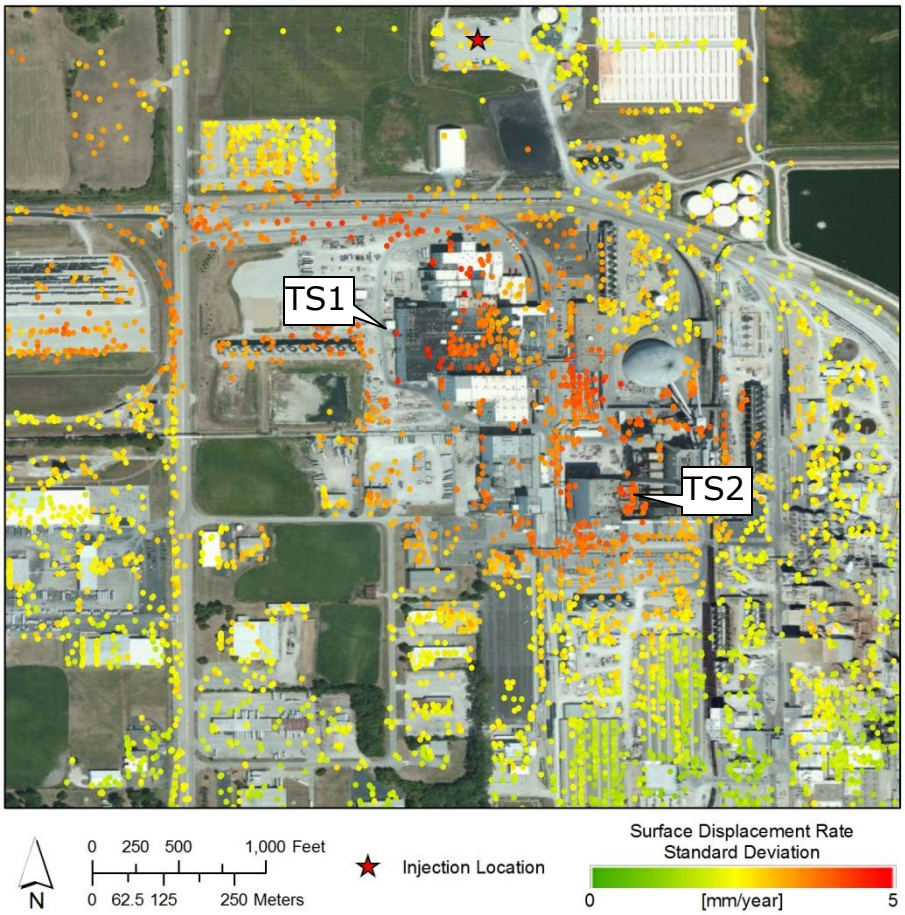


Figure 22. Standard deviation of the surface displacement rates identified over the area labelled Feature 3 in Figure 13.

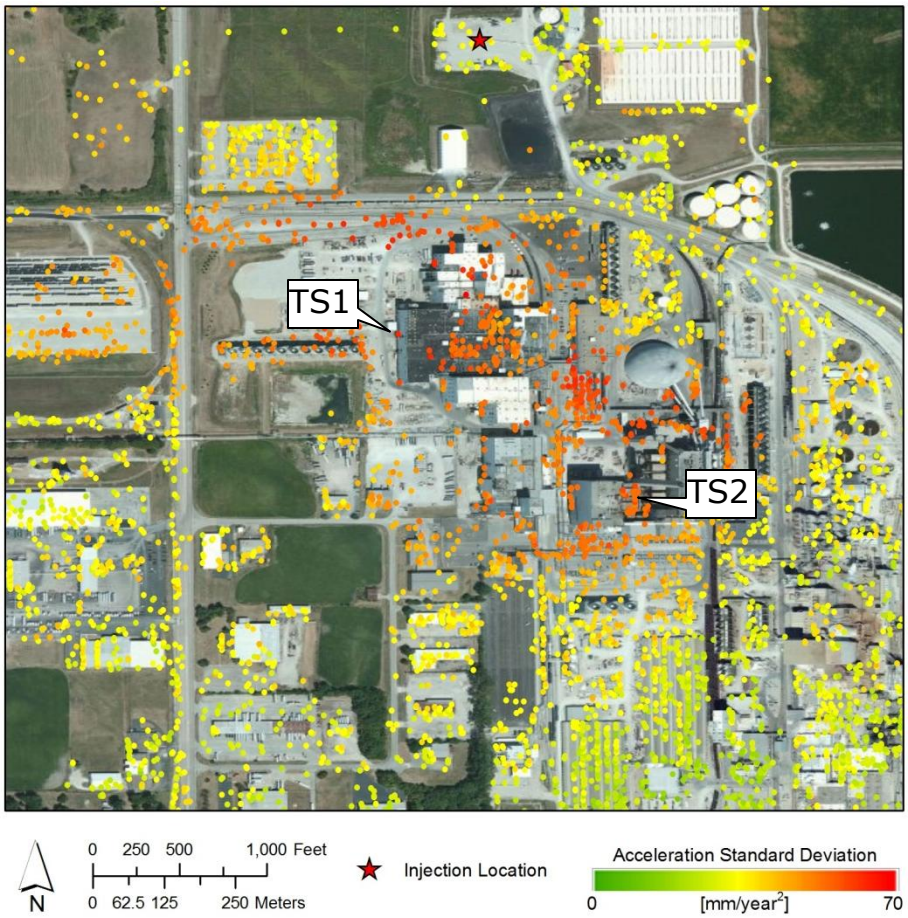


Figure 23. Standard deviation of the acceleration rates identified over the area labelled Feature 3 in Figure 13.

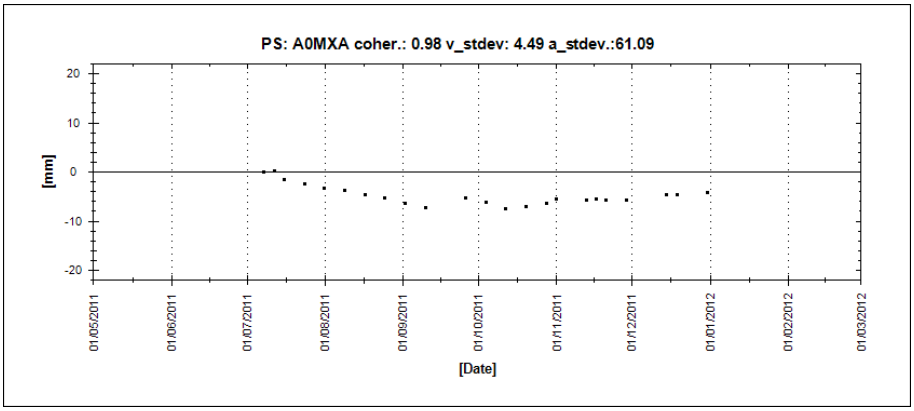


Figure 24. Time series for the point labelled TS1 in Figure 22 and Figure 23. Note the high displacement standard deviation (v_stdev) and acceleration standard deviation (a_stdev) values reported in the graph header.

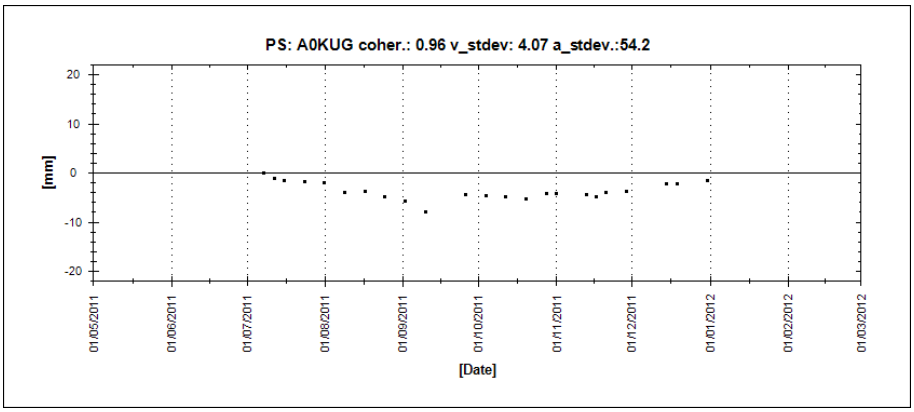


Figure 25. Time series for the point labelled TS2 in Figure 22 and Figure 23. Note the high displacement standard deviation (v_stdev) and acceleration standard deviation (a_stdev) values reported in the graph header.

5.4 Artificial reflector analysis

The locations and measured displacement of the artificial reflectors installed between the injector and the observation well are shown below (Figure 26).

The reflectors displayed little displacement (16 of 21 reflectors had displacement rates that were less than their associated standard deviation). This trend was expected considering the small overlap between the start of injection activities and the time period covered by the radar imagery. Movement was minimal for most of the reflectors over the analysis period and the measured displacement may be caused by seasonal changes in soil moisture, groundwater and/or temperature.

It is expected that with the continued acquisition of imagery over a longer time period, any seasonal effects will be characterized. Furthermore, the addition of imagery will also result in an increase in the precision of the measurements.

The time series of deformation for all 21 reflectors are shown in Appendix 2.

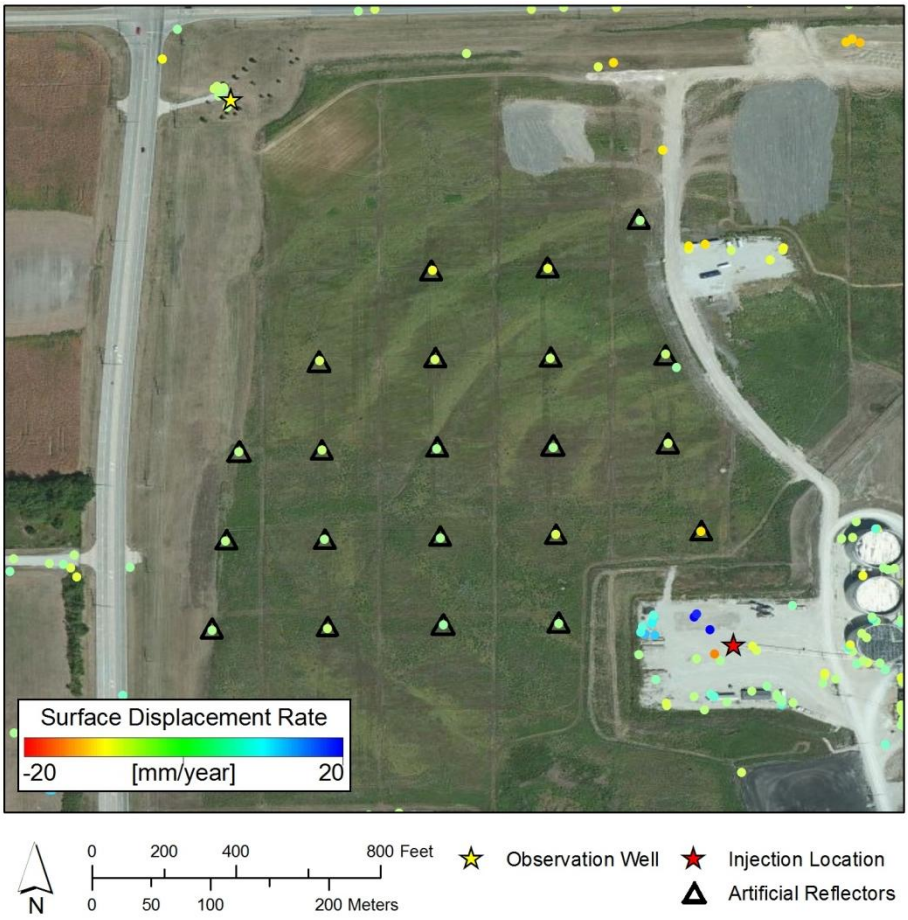


Figure 26. Surface displacement rates for the 21 artificial reflectors

5.5 Target distribution and density

A density of 2,738 points per square mile (1,057 points per square kilometer) was achieved from the processing of the first six months of radar imagery. This represents a more than 44% increase in the total number of measurement points identified from the feasibility study, which was 1,898 points per square mile (733 points per square kilometer). A summary of the points identified for these two analyses is shown below (Table 2). Statistics summarizing the results of the data processing can be found in Appendix 1.

Over 48% of the total number of measured points identified in the Cosmo-SkyMed dataset were of the distributed scatterer type (DS), accounting for nearly half of all targets identified. This demonstrates the usefulness of distributed scatterer approaches such as SqueeSAR™, which identify a much higher number of points than would have been identified using conventional permanent scatterer interferometry.

Number of points Identified	Feasibility Study	First Processing
Permanent Scatterers	5,822	13,822
Distributed Scatterers	12,680	12,873
Total	18,502	26,695

Table 2. Number of PS and DS candidate points identified in the feasibility study and the number of PS and DS measurement points identified in the present analysis.

The spatial coverage of radar targets varied throughout the AOI, and was primarily related to the presence of man-made structures including buildings, roads and other infrastructure. While small areas of exposed ground and several fields that do not appear to be actively worked produced some measurement points, any portions of the AOI where the surface is constantly changing such as in the case of active agriculture yielded few, if any, measurement points.

6 Summary

Uplift was observed in the area immediately surrounding the injection site, with rates ranging from 19.5±2.44 to 55.2±3.07 mm/year; however, the link to injection activities is not clear. Deformation rates over much of the area were affected by seasonal trends. There is no apparent correlation between the surface displacement data and CO_2 injection at this stage of the analysis.

Measurements from the artificial reflectors showed little movement based on the results of the current data processing. The only motion that was detected appears to be related to seasonal trends. At this point, there does not appear to be any correlation to the start of injection.

The first data processing and monitoring report over the Decatur injection site yielded a high density of measurement point across the AOI, enabling a good coverage of surface displacement information over the Decatur site. Distributed scatterers constituted approximately 50% of the identified radar targets.

7 Delivery of Data

The deliverables of the SqueeSAR™ analysis include the present report, the PS/DS data files and a software tool for assisting with the loading, viewing and interrogation of the data in ESRI ArcGIS 9.x software. The files contained on the accompanying CD are listed below (Table 3).

File name	Description
DECATUR_CSK_H4_05_A_T169-TSR.dbf	Table containing the height, velocity, velocity standard deviation, acceleration, coherence and time series of all the PS/DS identified in the analysis.
DECATUR_CSK_H4_05_A_T169-TSR.shp	ESRI Shapefile for displaying the database (dbf) file in a GIS environment.
DECATUR_CSK_H4_05_A_T169-REF.dbf	The reference point used for the ascending and descending data processing.
DECATUR_CSK_H4_05_A_T169-REF.shp	ESRI Shapefile for displaying the reference point file in a GIS environment.
Decatur.mxd	ESRI ArcGIS 9.3 project containing all PS data and the AOI shapefile.

Table 3. List of delivered files.

The ESRI ArcGIS 9.x project file is included to make it easier to view the data. Once the data contained in the CD have been saved to the user's hard drive it will be sufficient to open the project file with ArcGIS and update the links to indicate the new locations of the data. The project also contains various layers, including the reference point, artificial reflectors and the AOI shapefile.

8 The structure of the DBF files

The attributes of each PS/DS contained within the database are shown below (Table 4).

Field	Description
CODE	Unique identification code.
SHAPE	Indicates type of geometry (point).
HEIGHT [m]	Elevation above sea level of the PS and DS.
H_STDEV [m]	Standard deviation of PS and DS elevation value.
VEL [mm/yr]	PS movement rate. Positive values correspond to movement *toward the satellite (uplift);* negative values correspond to motion *away from the satellite (subsidence)*.
V_STDEV [mm/yr]	Standard deviation of PS and DS deformation rate.
ACC [mm/yr^2]	PS and DS acceleration rate.
A_STDEV [mm/yr^2]	Standard deviation of PS and DS acceleration value.
COHERENCE	Quality measure [between 0 and 1].
EFF_AREA [m^2]	Size of the area belonging to the PS and DS. For PS EFF_AREA = 0, for DS EFF_AREA > 0.
D(year/month/day) [mm]	Following the EFF_AREA column are a series of fields that contain the displacement values of successive acquisitions relative to the Master, expressed in mm.

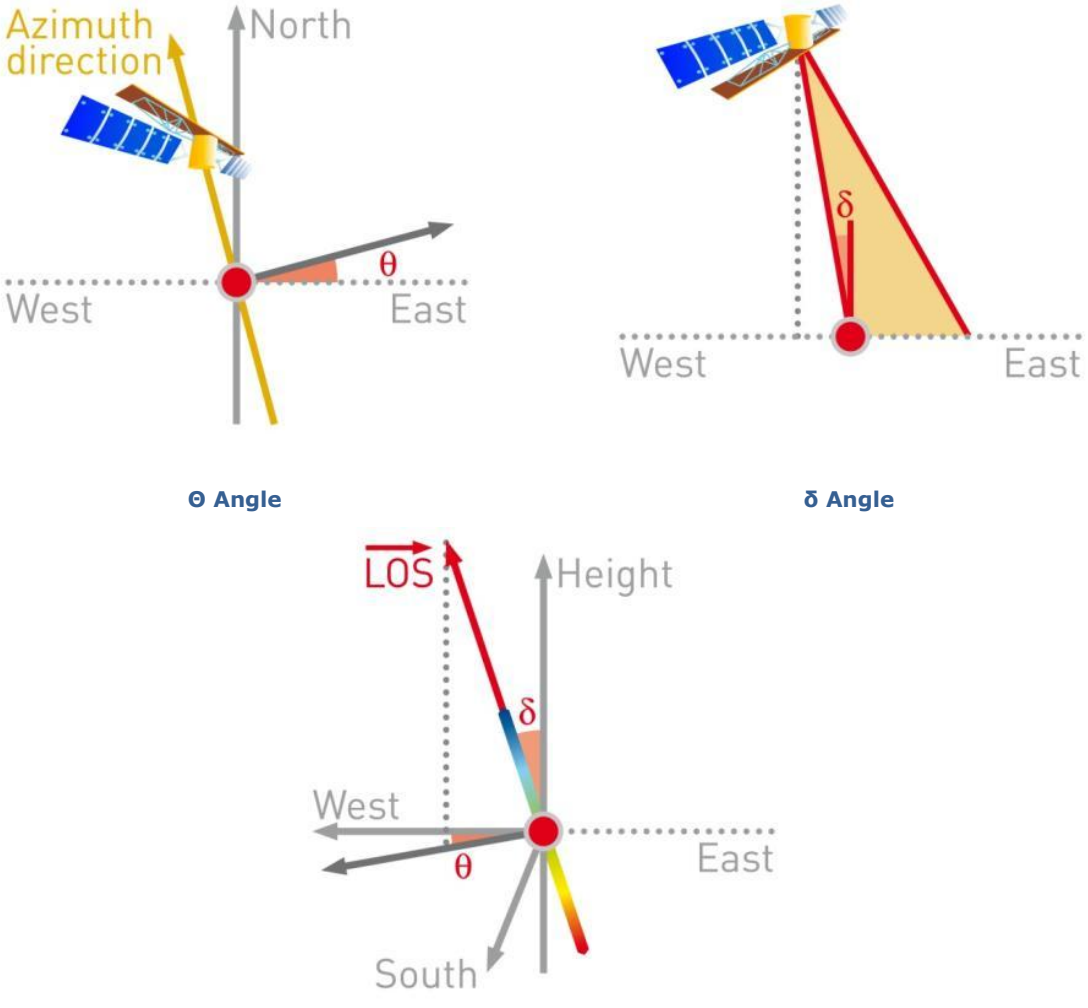
Table 4. Description of the fields contained in the Shapefile.

Appendix 1: Additional Properties of the SqueeSAR™ results

Radar data acquisition geometry

InSAR-based approaches measure surface displacement on a one-dimensional plane, along the satellite line-of-sight (LOS). The LOS angle varies depending on the satellite and on the acquisition parameters while the another important angle, that between the orbit direction and the geographic North, is nearly constant.

All Cosmo-SkyMed images used in the present analysis were acquired from an ascending orbit (travelling from south to north and imaging to the east) in beam mode H4-05. The symbol δ (delta) represents the angle the LOS forms with the vertical and Θ (theta) the angle formed with the geographic north (Figure 27). Table 5 lists the angles for this study.



Θ Angle **δ Angle**

Line Of Sight

Figure 27. Geometry of the image acquisitions over the Decatur site. θ represents the angle formed by the satellite with the north. δ represents the off-nadir angle formed by the beam.

Orbit geometry	Symbol	Angle
Ascending	**δ**	33.5
Ascending	**θ**	9.32

Table 5. Satellite viewing angles for the Cosmo-SkyMed imagery.

The line-of-sight (LOS) versor properties for the present study (Table 6). These values can be used to determine the sensitivity of the LOS to the vertical, E-W and N-S directions. Larger versor values imply a greater sensitivity to motion in that direction. Low versor values indicate that movement occurring in this direction will be more challenging to detect.

Direction	Versor
North	-0.0894
East	-0.5447
Vertical	0.8338

Table 6. Components of the LOS versors for the Cosmo-SkyMed data used in this study.

Data Processing

Both permanent scatterers (PS) and distributed scatterers (DS) were identified throughout the Decatur AOI. Buildings, roads, fences and other man-made structures provide the basis for many PS points in the present SqueeSAR™ analysis.

DS correspond to large areas (up to hundreds of square meters) and were identified from areas of infrastructure or anthropogenic features, bare or sparsely vegetated ground, exposed areas and fallow fields. It is important to consider that while DS are represented as individual points for clarity of presentation and ease of interpretation, these measurements actually correspond to non-point features that are multiple pixels in size. The size of the DS within the AOI range from 76 to 937 m^2 (corresponding to DS approximately 8 to 104 pixels in size).

A summary of the other properties relative to the data processing (Table 7).

Satellite	Cosmo-SkyMed
Acquisition geometry	Ascending
Critical baseline [m]	5734.12
Analysis time interval	08 July 2011 to 31 December 2011
Number of scenes processed	23
Georeferencing	PS aligned on Microsoft Virtual Earth
Projection system used / datum	NAD 1983 UTM Zone 16N
Reference Point location	4413928.7N; 338071.1E
Area of Interest	9.75 miles squared (25.25 km^2)
Number of PS + DS identified - **Number of PS** - **Number of DS**	26,695 13,822 12,873
Number of Time Series	26,695
Average PS + DS density	2,738 PS and DS per square mile (1,057 PS and DS/km^2)

Table 7. Statistics of the ascending Cosmo-SkyMed processed data.

The critical baseline is the maximum admissible separation distance between orbits for an InSAR analysis. Interferometry cannot be performed if the orbits are separated by more than this distance.

Precision Assessment

Three parameters are used to characterize the results:

- Precision of the estimated deformation rates;
- Precision of the estimated elevations;
- Precision of the geocoding.

The typical precision values applicable to PS located within 2 km from the reference point when **at least 45 radar images** have been processed (Table 8).

DEFORMATION RATE	< 1 [mm/yr]
DISPLACEMENT ERROR (single displacement between contiguous satellite images)	< 5 [mm]
ELEVATION	± 1.5 [m]
POSITIONING ERROR ALONG EAST DIRECTION	± 3 [m]
POSITIONING ERROR ALONG NORTH DIRECTION	± 2 [m]

Table 8. Measurement accuracies for PS located within 2 km of the reference point, based on the processing of at least 45 SAR images.

Appendix 2: Artificial Reflector Displacement Measurements

Displacement results for the 21 artificial reflectors installed at the Decatur site (Figure 28) are shown below (Figure 29 to Figure 49, inclusive).

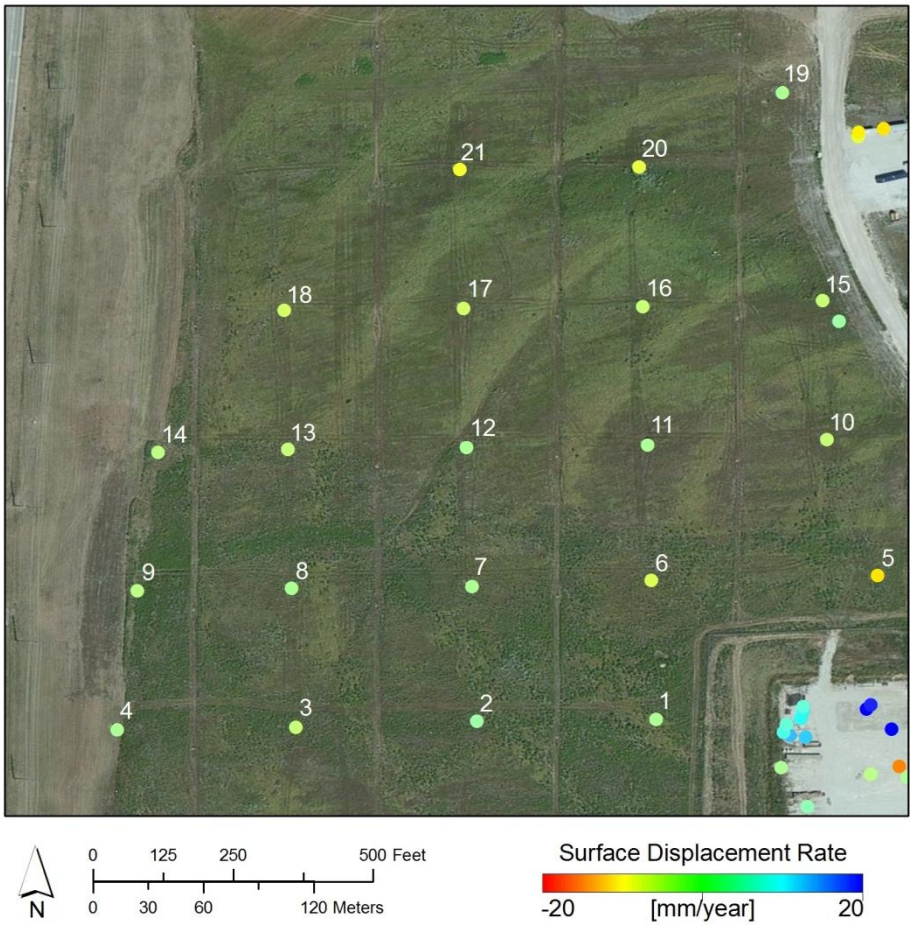


Figure 28. Artificial reflectors installed at the Decatur site, each labeled with a corresponding time series shown in Figure 29 to Figure 49.

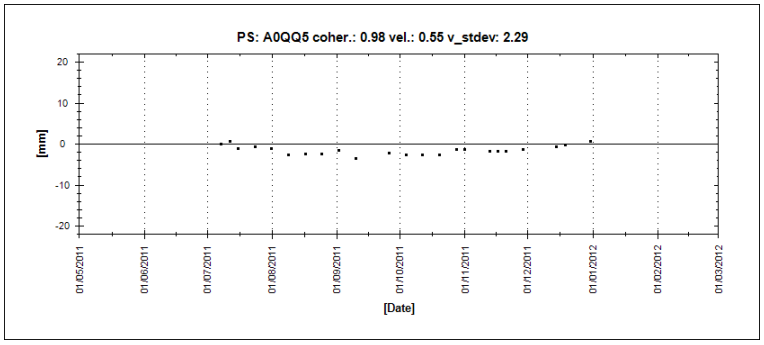


Figure 29. Time series for the artificial reflector labeled 1 in Figure 28.

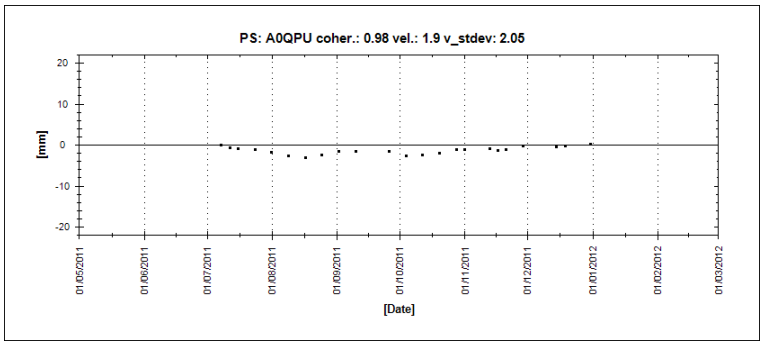


Figure 30. Time series for the artificial reflector labeled 2 in Figure 28.

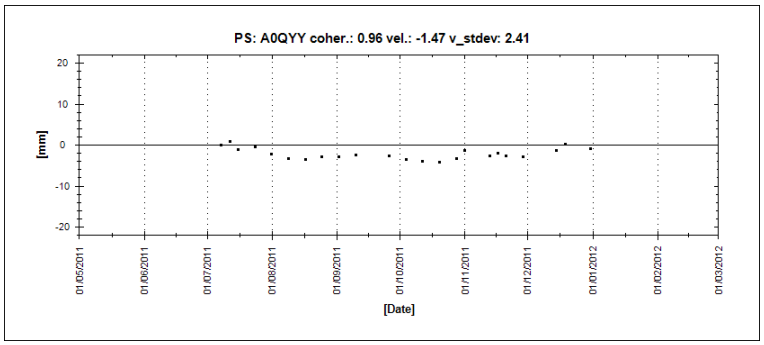


Figure 31. Time series for the artificial reflector labeled 3 in Figure 28.

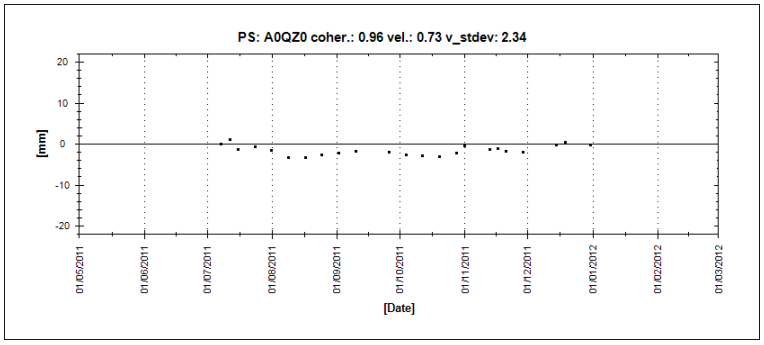


Figure 32. Time series for the artificial reflector labeled 4 in Figure 28.

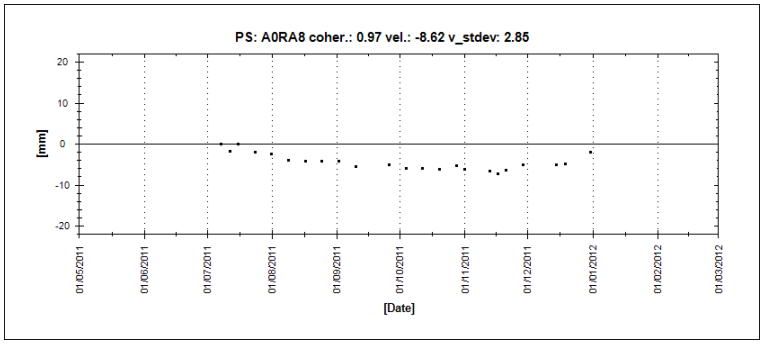


Figure 33. Time series for the artificial reflector labeled 5 in Figure 28.

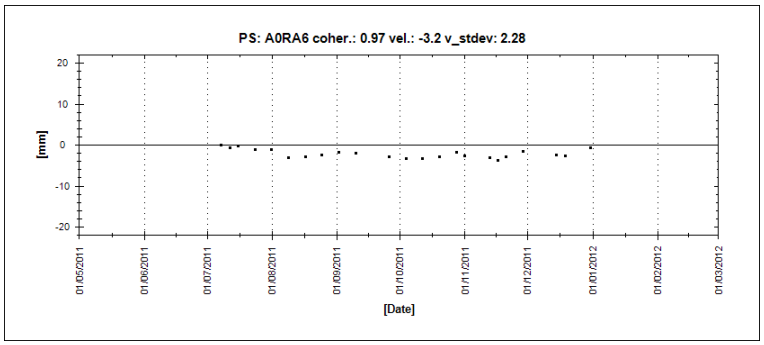


Figure 34. Time series for the artificial reflector labeled 6 in Figure 28.

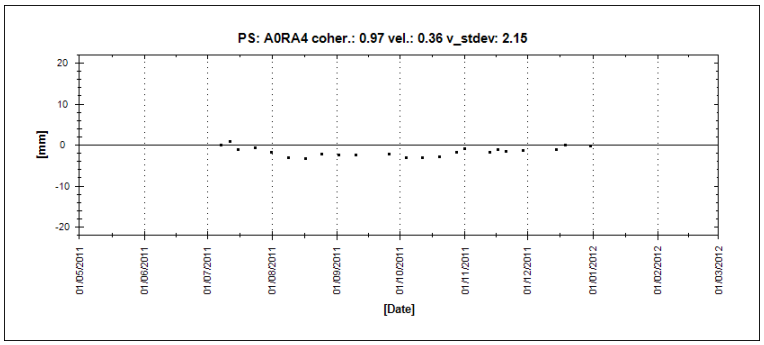


Figure 35. Time series for the artificial reflector labeled 7 in Figure 28.

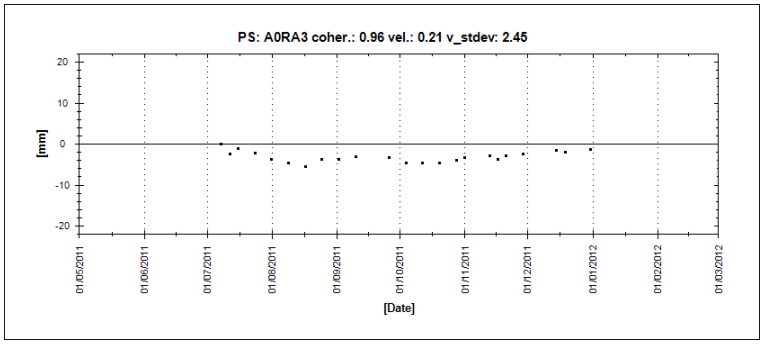


Figure 36. Time series for the artificial reflector labeled 8 in Figure 28.

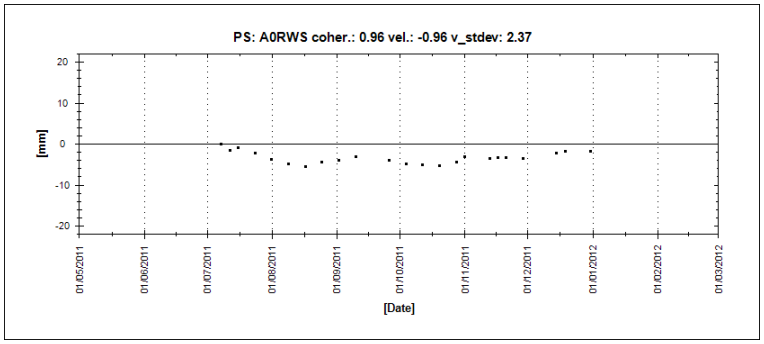


Figure 37. Time series for the artificial reflector labeled 9 in Figure 28.

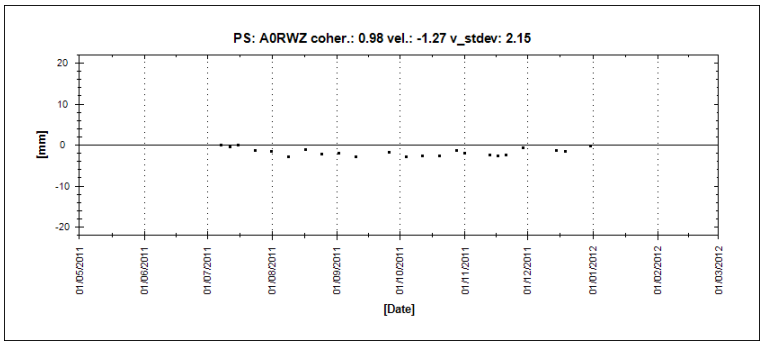


Figure 38. Time series for the artificial reflector labeled 10 in Figure 28.

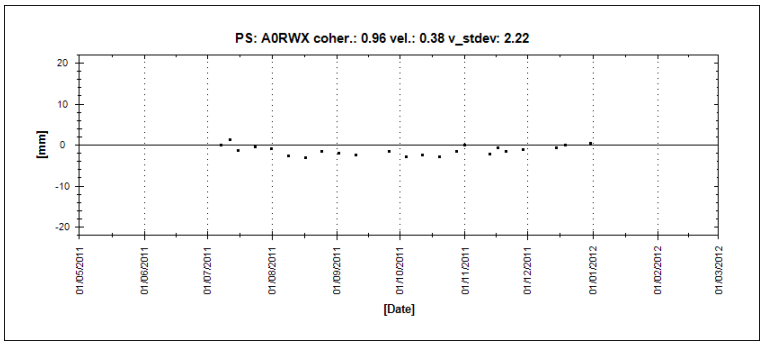


Figure 39. Time series for the artificial reflector labeled 11 in Figure 28.

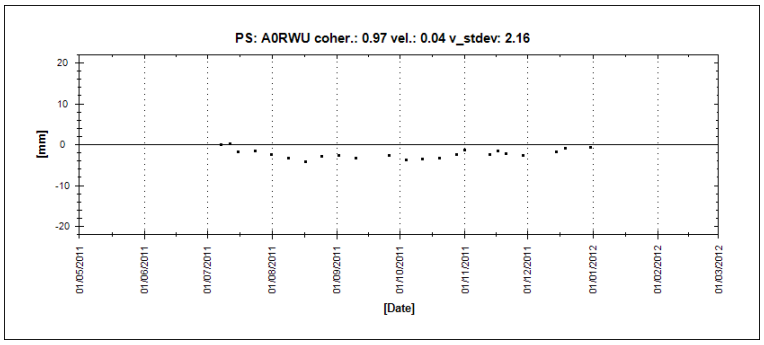


Figure 40. Time series for the artificial reflector labeled 12 in Figure 28.

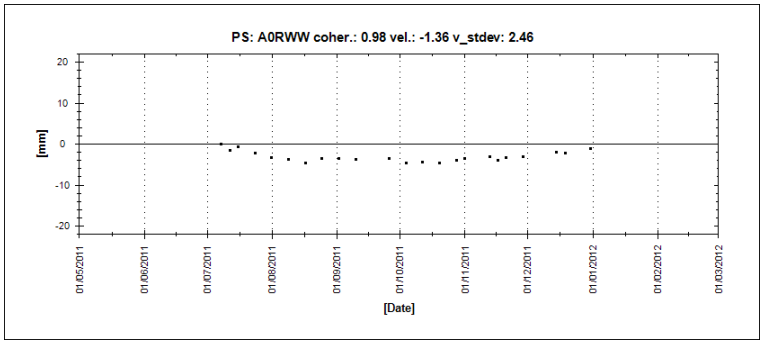


Figure 41. Time series for the artificial reflector labeled 13 in Figure 28.

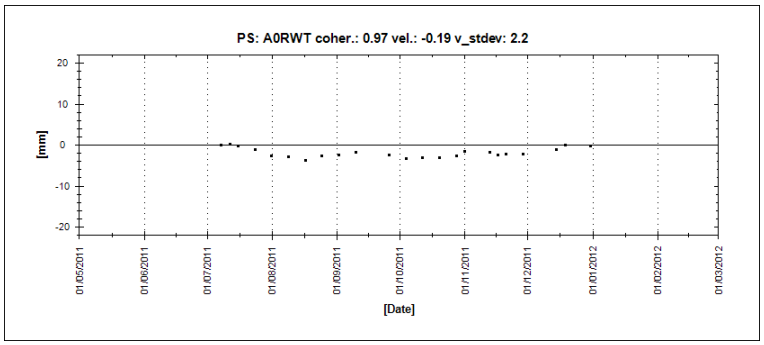


Figure 42. Time series for the artificial reflector labeled 14 in Figure 28.

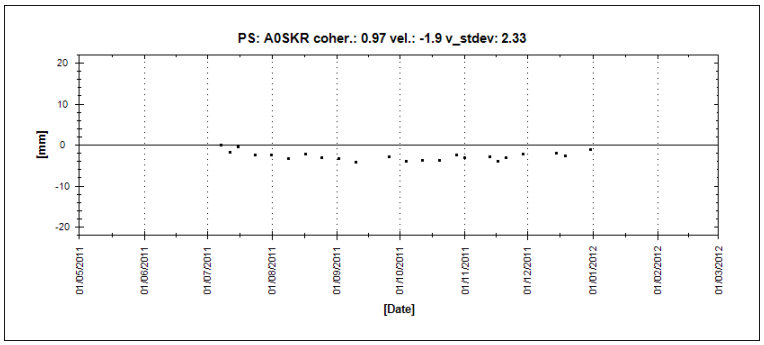


Figure 43. Time series for the artificial reflector labeled 15 in Figure 28.

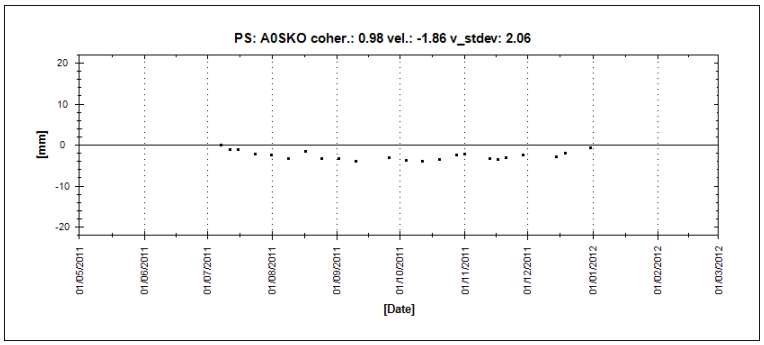


Figure 44. Time series for the artificial reflector labeled 16 in Figure 28.

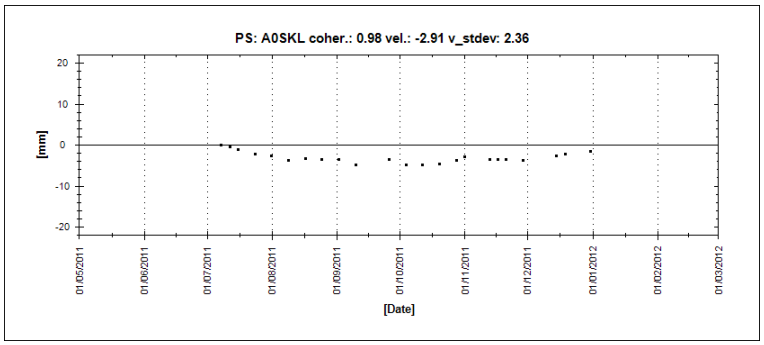


Figure 45. Time series for the artificial reflector labeled 17 in Figure 28.

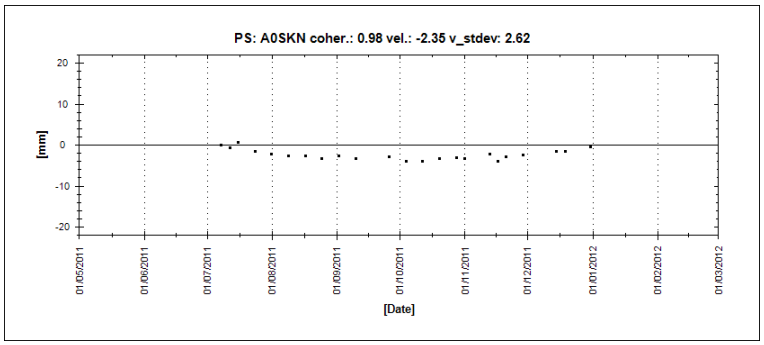


Figure 46. Time series for the artificial reflector labeled 18 in Figure 28.

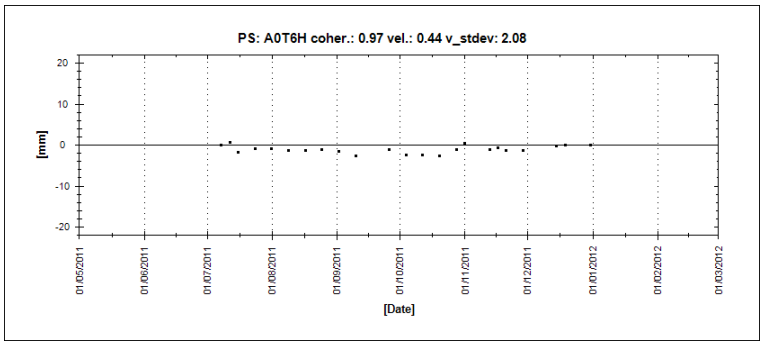


Figure 47. Time series for the artificial reflector labeled 19 in Figure 28.

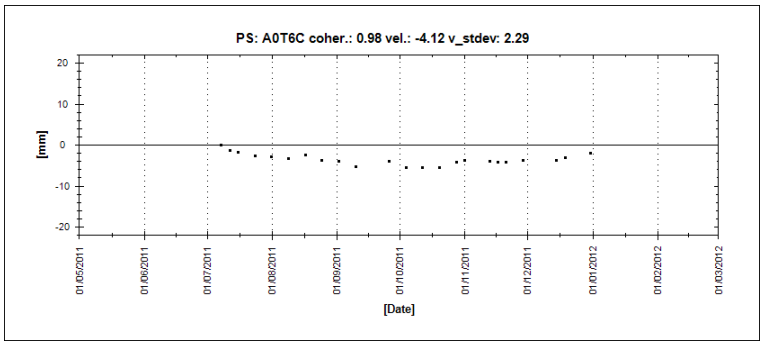


Figure 48. Time series for the artificial reflector labeled 20 in Figure 28.

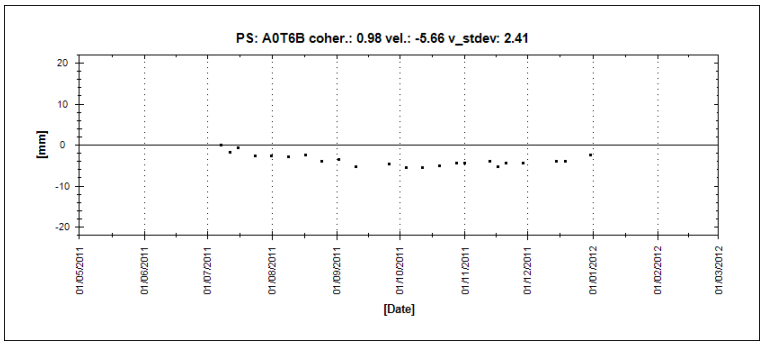


Figure 49. Time series for the artificial reflector labeled 21 in Figure 28.

Appendix 3: SqueeSAR™ Processing

InSAR

Interferometric Synthetic Aperture Radar, also referred to as SAR interferometry or InSAR, is the measurement of signal phase change (interference) between radar images. When a point on the ground moves, the distance between the sensor and the point changes, thereby producing a corresponding shift in signal phase. This shift is used to quantify the ground movement.

An interferogram is a 2D representation of the difference in phase values. Variations of phase in an interferogram are identified by fringes, colored bands that indicate areas where and how much movement is occurring. The precision with which the movement can be measured is usually in the centimeter range as the phase shift is also impacted by topographic distortions, atmospheric effects, and other sources of noise.

DInSAR

When InSAR is used to identify and quantify ground movement the process is referred to as Differential InSAR (DInSAR). In DInSAR topographic effects are removed by using a DEM of the area of interest to create a differential interferogram.

DInSAR is still impacted by atmospheric effects, as there is no method for removing this signal phase contribution. It is a useful tool for identifying footprints of progressing movement and creating deformation maps. The limitations of DInSAR are its relatively low precision (centimeter scale) and that it cannot distinguish between linear and non-linear motion.

PSInSAR™

PSInSAR™ is an advanced form of DInSAR. The fundamental difference is that it uses multiple interferograms created from a stack of at least 15 radar images.

PSInSAR™ was developed to overcome the errors produced by atmospheric artifacts on signal phase. The PSInSAR™ algorithm automatically searches the interferograms for pixels that display stable radar reflectivity characteristics throughout every image of the dataset. In PSInSAR™ these pixels are referred to as Permanent Scatterers (PS). The result is the identification of a sparse grid of point-like targets on which an atmospheric correction procedure can be performed. Once these errors are removed, a history of motion can be created for each target, allowing the detection of both linear and non-linear motion.

The result is a sparse grid of PS that are color-coded according to their deformation rate and direction of movement. The information available for each PS includes its deformation rate, acceleration, total deformation, elevation, coherence as well as a time series of movement. PSInSAR™ measures ground movement with millimeter accuracy.

SqueeSAR™

PS are objects, such as buildings, fences, lampposts, transmission towers, crash barriers, rocky outcrops, etc, that are excellent reflectors of radar microwaves. However, TRE has noticed that many other signals are present in the processed data. These do not produce the same high signal-to-noise ratios of PS but are nonetheless distinguishable from the background noise. Upon further investigation it was found that the signals are reflected from extensive homogeneous areas where the back-scattered energy is less strong, but statistically consistent. These areas have been called distributed scatterers (DS) and correspond to rangeland, pastures, bare earth, scree, debris fields, arid environments, etc (Figure 50).

The SqueeSAR™ algorithm was developed to process the signals reflected from these areas. As SqueeSAR™ incorporates PSInSAR™ no information is lost and movement measurement accuracy is unchanged.

SqueeSAR™ also produces improvements in the quality of the displacement time series. The homogeneous areas that produce DS normally comprise several pixels. The single time series attributed to each DS is estimated by averaging the time series of all pixels within the DS, effectively reducing noise in the data.

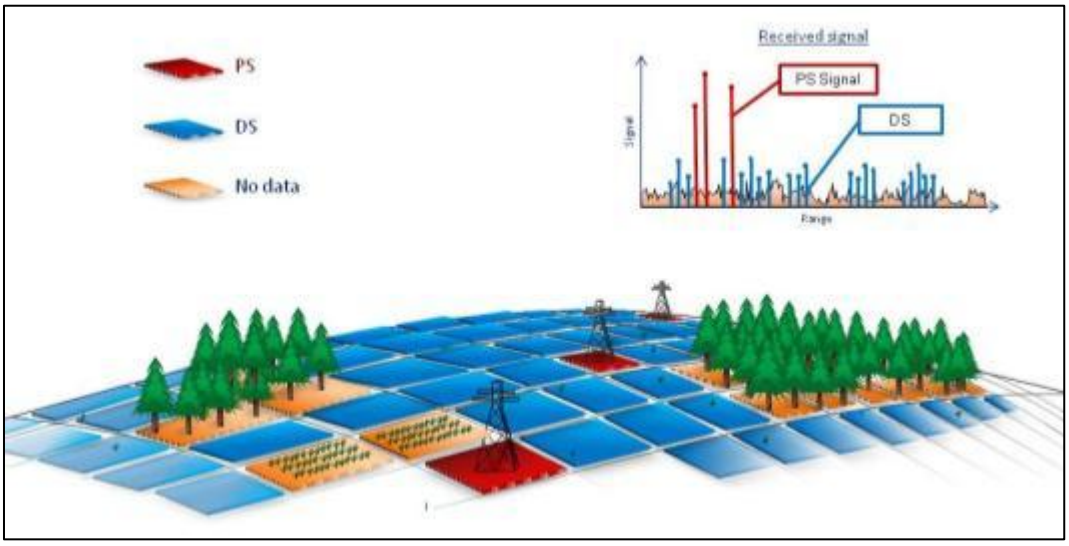


Figure 50. Illustration of the identification of permanent (PS) and distributed scatterers (DS) by the SqueeSAR™ algorithm.

Appendix 4: Data processing

Methodology

The identification of PS and DS in a series of radar images comprises a sequence of steps.

First, all radar data archives are screened to determine the most suitable source of raw data for the particular area of interest and to select all the high quality images within the chosen dataset.

As the signal echo from a single point target contains many returning radar pulses it appears defocused in a synthetic aperture radar (SAR) raw image. The first processing step is therefore to focus all the received energy from a target in one pixel. The images are then precisely aligned to each other, or co-registered, and analyzed for their suitability for interferometry. The parameters that are analyzed are the normal baseline and the temporal distribution of the images.

There then follows a number of statistical analyses on the phase and amplitude characteristics of the backscattered radar signal that return to the satellite. If a concentrated number of signals reflect off a particular feature within a pixel and backscatter to the satellite, the feature is referred to as a 'scatterer'. When the same scatterer appears in all, or most, of a data set of SAR images of a particular location, then the scatterer is deemed to be 'permanent'.

At this stage it is possible to identify a subset of pixels, referred to as Permanent Scatterer Candidates (PSC), that are used to estimate the impact on signal phase of ionospheric, tropospheric and atmospheric effects, as well as possible orbit errors.

Once the signal phase has been corrected for these effects, any remaining changes in signal phase directly reflect ground movement.

Master Image Selection

SqueeSAR™ requires that one image (or scene) in each dataset has to become both a geometric and temporal reference to which all the other images are then related. This image is referred to as the master image and those that remain are slave images.

The master image should be chosen according to the following criteria:

- it minimizes the spread of normal baseline values for the slave images;
- similarly, it minimizes the temporal baseline values between the master and each slave image; and
- it minimizes the effects of signal noise arising from changes in vegetation cover and/or small changes in the look angle of the satellite from one scene to another.

Signal Phase and Amplitude Analysis

General

Each pixel of a SAR image contains information on the amplitude of signals that are backscattered toward the satellite, as well as on the signal phase. The amplitude is a measure of the amount of the radar pulse energy reflected, while the phase is related to the length of the path of the electromagnetic wave, from the platform to the ground and back again.

Analyses of both amplitude and phase of the SAR image provide an indication of the stability of each pixel, over time, whereby it is possible to identify those pixels that are most likely to behave as Permanent Scatterers. Statistical methods are used extensively in this process.

Among the different statistical parameters that can be computed two are of particular interest: the Phase Stability Index (PSI), obtained from the phases of the images within the dataset, and the Multi Image Reflectivity (MIR) map, derived from the amplitude values of the available acquisitions.

Radar Phase and Coherence

The phase stability is strongly linked to the concept of coherence. Pixels that consistently display high phase stability are said to be coherent. Coherence is measured by an index that ranges from 0 to 1. When a pixel is completely coherent, it will have a coherence value of 1. Correspondingly, if a pixel has a low phase stability, its coherence index will be 0. In general, interferometry is successful when the coherence index lies between 0.5 and 1.0.

Radar Amplitude and Multi-Image Reflectivity

The amplitude of a pixel within a SAR image is the aggregate of the backscattered energy toward the satellite from within the pixel's equivalent land area. This equivalent land area is referred to as the radar resolution, and in the case of the Cosmo-SkyMed satellite, it measures about 3 by 3 meters. It is necessary to look into the amplitude values of all the images in the dataset, in order to understand exactly what was seen by the satellite at the time of each acquisition.

If a target has experienced significant change in its surface characteristics it will exhibit variation in its reflectivity (electromagnetic response) between two acquisitions. In such circumstances, the possibility of detecting movement by means of SAR interferometry is seriously compromised. The signal phase difference between the two images now contains not only the contribution due to displacement, but also that due to the change in the reflectivity of the target. This prevents, in the worst case, the obtaining of any useful information on ground movement.

Accordingly, it is necessary to look into the amplitude values of all the images in the dataset, in order to understand exactly what was seen by the satellite at the time of each acquisition.

Another artifact linked to amplitude is known as speckle. Speckle is random noise that appears as a grainy salt and pepper texture in an amplitude image. This is caused by random interference from the multiple scattering returns that occur within each resolution cell. Speckle has an adverse impact on the quality and usefulness of SAR images. However, the higher the number of images taken of the same area at different times or from slightly different 'look' angles, the easier it is to reduce speckle. This increases the quality and level of details of the amplitude image enabling it to be used as a background layer for observing the presence of PS results.

The Multi Image Reflectivity (MIR) map is the means by which speckle reduction is accomplished. Averaging a number of images tends to negate the random amplitude variability, leaving the uniform amplitude level unchanged (Figure 51).

It should be emphasized that the information in the MIR map is the reflectivity of each pixel, i.e. the ability to backscatter the incident wave toward the satellite. Flat surfaces (roads, highway, rivers, lakes) act like a mirror, meaning that if their orientation is not exactly perpendicular to the incident wave negligible energy is reflected back to the

sensor; they appear dark in the image. On the other hand, because of their irregular physical shape, metal structures or buildings reflect a significant portion of the incident signal back to the radar, resulting in very bright pixels in the MIR map.

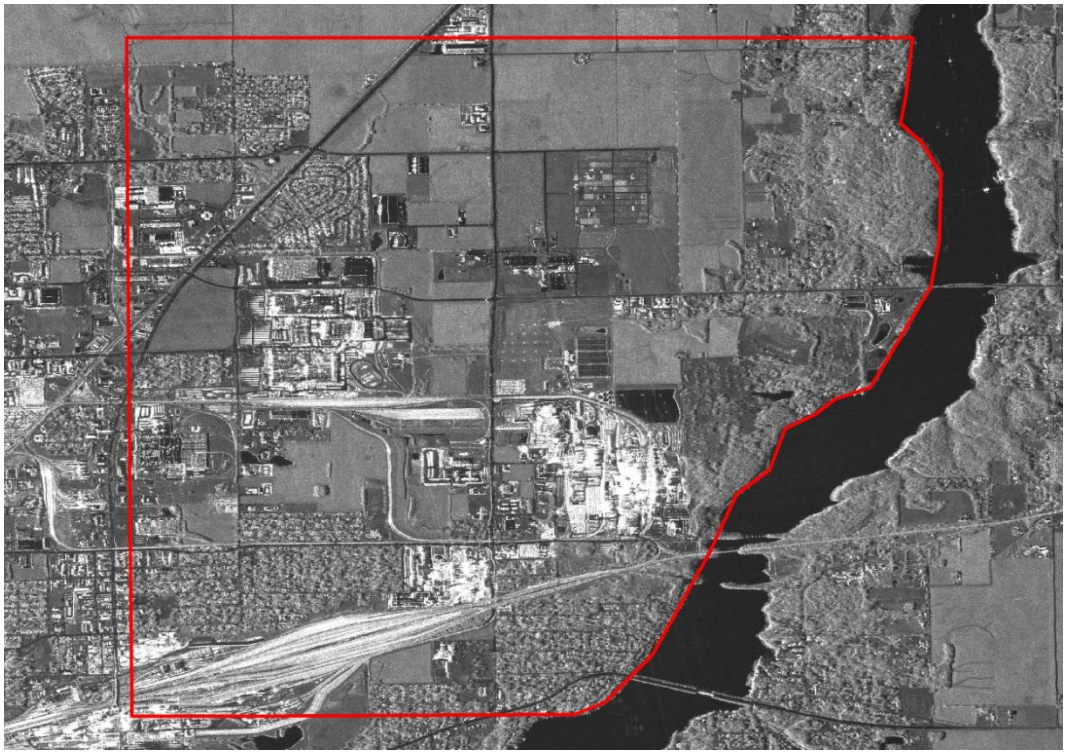


Figure 51. Multi-image reflectivity map of the ascending Cosmo-SkyMed dataset over the Decatur site.

Interferograms

After the statistical analyses of the SAR images have been completed, a set of differential interferograms is generated. This entails subtracting the phase of each slave image from the phase of the master image. In doing so, the difference in signal path length between the two images is calculated. This difference is related to possible ground motion.

In any SAR image, there are embedded topographic distortions that arise during image acquisition. These are removed using a reference Digital Elevation Model (DEM), leaving ground movement and the signal phase distortions arising from atmospheric effects as the only embedded variables.

The differential interferograms represent the starting point for applying the PSInSAR™ approach.

Estimation of the atmospheric effects

When a radar signal enters and exits a moisture-bearing layer in the atmosphere, its wavelength can be affected, introducing potential errors into the signal path length. The removal of atmospheric impacts is fundamental for increasing the precision of ground movement measurement.

A sub-set of pixels, usually corresponding to buildings, lampposts, antennas, small structures and exposed rocks, is chosen from among those that have high PSI values.

These are referred to as PS Candidates (PSC). PSC density is, of course, higher in towns and cities rather than in forests and vegetated areas. However, it is often possible to obtain good PSC density in rural areas.

For each image, the atmospheric impacts are estimated at each PSC location. The process is statistically based and benefits in accuracy by the greater the number of available images for the analysis. By comparing the atmospheric contribution on neighboring pixels that would be experiencing the same atmospheric conditions, the atmospheric contribution can be reconstructed over the whole image.

The processed dataset allows identification of a PSC cluster dense enough to identify and extract the atmospheric contribution over the entire area of interest.

Post-processing

In this stage the processed data undergoes a thorough quality control following ISO 9001:2000 guidelines. The PS data is checked for anomalies, aligned on an optical image layer usually provided by the client and the final report is prepared.

Appendix 4: Abbreviations and Acronyms

APS	Atmospheric Phase Screen
ASAR	Advanced Synthetic Aperture Radar
AOI	Area Of Interest
DEM	Digital Elevation Model
DInSAR	Differential Interferometric SAR
ERS	Earth Resources Satellite
ESA	European Space Agency
FAR	False Alarm Rate
GCP	Ground Control Point
GIS	Geographic Information System
GPS	Global Positioning System
InSAR	Interferometric SAR
LOS	Line Of Sight
ML	Maximum Likelihood
MLS	Minimum Least Squares
POLIMI	Politecnico di Milano
PS	Permanent Scatterer(s)
PSI	Phase Stability Index
PSInSAR™	Permanent Scatterers SAR Interferometry is a world-wide POLIMI Trademark
SAR	Synthetic Aperture Radar
SNR	Signal to Noise Ratio
SqueeSAR™	The most recent InSAR algorithm patented by TRE
TRE	Comprehensive term for Tele-Rilevamento Europa and TRE Canada
TS	(Permanent Scatterer Displacement) Time Series



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